Overprecision and Subjectively Rational Inattention*

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Abstract

We incorporate overprecision—the tendency to overestimate the precision of one's beliefs—into a canonical rational inattention model. In the benchmark model, overprecision reduces responsiveness to new information and can amplify the effect of processing costs on belief updating. We test these predictions in an incentivized online experiment, independently varying calibration feedback and information complexity. Feedback increases reported uncertainty and suggests greater responsiveness among initially overprecise participants, with less clear effects among initially underprecise participants. Greater complexity reduces responsiveness and posterior accuracy. However, we do not detect the model's predicted amplification effect. We explore possible explanations for this null result.

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1 Introduction

The canonical full-information rational-expectations benchmark assumes that agents incorporate all available information when forming beliefs. Empirical evidence often points to incomplete responses to available information. For example, households tend to underreact to changes in taxation and to monetary-policy announcements (Chetty et al., 2009; Taubinsky and Rees-Jones, 2018; Coibion et al., 2020, 2021). A prominent explanation for such underreaction is rational inattention (Sims, 2003). At the core of this theory is a simple premise: processing information is costly. If the expected benefit of processing information does not exceed its cost, ignoring that information can be rational (Woodford, 2001; Sims, 2003; Maćkowiak et al., 2023).

While the assumption that agents can assess the costs and benefits of information processing without bias is a useful benchmark, it is unlikely to hold in general. A large body of work documents systematic departures from frictionless Bayesian information processing, including base-rate neglect (Kahneman and Tversky, 1973; Tversky and Kahneman, 1974), correlation neglect (Enke and Zimmermann, 2019), complexity effects (Oprea, 2020), and limited cognitive ability (D'Acunto et al., 2021). More recently, the literature has highlighted overprecision as a systematic and pervasive bias shaping various aspects of decision-making and behavior (Kahneman, 2011; Moore et al., 2015; Bosch-Rosa et al., 2024).

Overprecision is a form of overconfidence arising from a metacognitive bias whereby individuals systematically overestimate the precision of their information (Moore, 2022). Relative to Bayesian agents with calibrated beliefs about their uncertainty, overprecise agents therefore perceive less scope for learning from new information. This miscalibration can distort the cost-benefit trade-off that governs attention allocation.

In this paper, we examine theoretically and empirically how overprecision influences attention and shapes information use. Our model distinguishes two effects of overprecision on updating: it reduces the weight placed on new information at a given level of attention, and, by lowering the perceived gains from processing new signals, it can also reduce the attention devoted to them. We refer to this mechanism as *subjectively rational inattention*. Agents choose attention optimally given their perceived prior precision and the cost of attention. Their behavior is therefore subjectively rational, even though their perceived precision is objectively miscalibrated.

We formalize this mechanism in a Gaussian updating model with endogenous attention. An agent has a prior over a continuous state and receives a noisy public signal. By paying attention, the agent can reduce the additional noise introduced when processing the signal. Overprecision enters as a wedge between calibrated prior-error variance and perceived prior variance. The model delivers three predictions. First, holding attention costs fixed, higher overprecision lowers the weight placed on the signal. Second, holding perceived prior precision fixed, higher processing costs lower the signal weight. Third, when the relevant attention choices are interior, overprecision amplifies the effect of attention costs: a cost increase reduces updating more sharply when the agent perceives the prior as more precise.

We test these predictions using a novel task embedded in a pre-registered, incentivized online experiment with a representative German sample. The survey consists of two blocks. In the first block, participants view a series of pictures depicting groups of people and estimate the average

age of each group at the time the picture was taken. For each picture, we also elicit participants' expected absolute error. Following the Subjective Error Method of Bosch-Rosa et al. (2024), we measure overprecision as the difference between realized and expected absolute errors. In the second block, participants again provide age estimates and expected errors for a new set of pictures. They can then update each estimate after viewing a signal: a cloud-style display containing a random subset of ages from the picture.

The experiment manipulates the model's two key components. To change how participants perceive the accuracy of their prior beliefs, we provide feedback on whether their earlier estimates were farther from or closer to the truth than they had expected. To vary processing costs, we change signal complexity. In Easy-Info, participants see only the relevant ages; in Hard-Info, the same ages appear in the same positions but are surrounded by irrelevant German nouns. The signal therefore contains the same information in both conditions but is harder to process in Hard-Info.

The design has several useful features. First, it measures overprecision without asking respondents to provide confidence intervals, which many subjects find unfamiliar (Teigen and Jørgensen, 2005; Moore et al., 2015). Second, the decoy-word treatment provides a simple manipulation of processing complexity while holding the signal fixed. Third, the continuous state variable lets us distinguish the mean of beliefs from their perceived precision, which is essential for studying overprecision rather than only biased prior means.

Among initially overprecise participants, feedback increases reported uncertainty and provides suggestive evidence of greater responsiveness to new information. We find no similarly clear response among initially underprecise participants. Making the signal harder to process also reduced the responsiveness to the displayed information and left final beliefs less accurate. The key question, however, is whether these two forces reinforce one another. The estimated interactions are statistically indistinguishable from zero, so the experiment does not establish the predicted amplification effect.

To further investigate this null result, we go beyond the pre-registered analysis and consider statistical, theoretical, and design-based explanations for the absence of the interaction. An exploratory analysis using measured overprecision yields negative interactions with information complexity, but the estimates are correlational and not robustly statistically significant. These analyses help characterize the null without changing the conclusion from the randomized treatment test.

We contribute to the existing literature in three ways. First, we study how misperceived prior precision interacts with information-processing costs, contributing to research on behavioral influences on attention (Gabaix, 2019; Maćkowiak et al., 2023). Related work examines how news utility or emotional payoffs shape attention (Pagel, 2018; Bolte and Raymond, 2024). Our model provides a complementary metacognitive mechanism: agents choose attention to improve forecast accuracy, but misperceive the precision of their prior knowledge. Second, we extend research on feedback and overprecision (Moore and Healy, 2008; Moore et al., 2015) by examining the consequences of calibration feedback for subsequent information use. Among initially overprecise participants, feedback increases reported uncertainty and provides suggestive

evidence of greater responsiveness to new information. Because feedback may affect updating through several channels, these effects do not establish that reduced overprecision accounts for the additional updating. Third, we contribute to experimental research on attention allocation (Dean and Neligh, 2023; Bronchetti et al., 2023) by testing whether calibration feedback changes responsiveness to information complexity. Independently randomizing these two interventions allows us to examine the model's amplification prediction alongside their separate effects on updating.

Our analysis highlights how beliefs about one's own knowledge shape the perceived value of new information. The model shows that agents who overestimate their prior precision can choose attention optimally given their beliefs yet respond less to new information than a calibrated benchmark would predict. This perspective connects research on overconfidence with the economics of costly information processing and motivates evaluating calibration feedback alongside efforts to make information clearer and easier to process. In settings such as financial education and public policy communication, feedback about the accuracy of past judgments could help recipients make better use of subsequent information. Together, clearer information and calibration feedback would target both processing costs and recipients' perceptions of how much they stand to learn.

2 Model and hypotheses

2.1 Beliefs, overprecision, and signal processing

We start with a standard Bayesian updating framework in which agent i forms a belief about a continuous state θ. The agent's prior has mean $\mu_{\theta,i}$. From the analyst's perspective, the error in this prior mean has calibrated variance $\sigma^2_{\theta,i}$, whereas the agent perceives its variance to be $\widetilde{\sigma}^2_{\theta,i}$. Overprecision is captured by

$$\widetilde{\sigma}^2_{\theta,i} = \frac{\sigma^2_{\theta,i}}{\omega_i}, \tag{1}$$

where $\omega_i \in (0, \infty)$. If $\omega_i > 1$, the agent is overprecise and perceives the prior as more precise than warranted by its calibrated accuracy. If $\omega_i < 1$, the agent is underprecise and perceives the prior as less precise than warranted by its calibrated accuracy. This formulation is useful in our experimental setting because the state is continuous, which lets us separate the belief's location from its perceived precision.

The Gaussian benchmark also provides a direct link between the structural overprecision parameter ω_i and the error measures used in the experiment. For a zero-mean normal error, expected absolute error equals $\sqrt{2/\pi}$ times its standard deviation. Let $\kappa = \sqrt{2/\pi}$. We therefore define CAE_i as the expected absolute error implied by the calibrated distribution of prior errors, and SE_i as the expected absolute error perceived by the agent. Under the calibrated distribution $N(0, \sigma^2_{\theta,i})$ and the agent's perceived distribution $N(0, \sigma^2_{\theta,i}/\omega_i)$, these quantities are

$$CAE_i = \kappa\sigma_{\theta,i}, \qquad SE_i = \kappa\frac{\sigma_{\theta,i}}{\sqrt{\omega_i}}. \tag{2}$$

Hence, an agent is overprecise exactly when perceived expected error is smaller than calibrated expected error, $CAE_i - SE_i > 0$. This difference is the population analogue of the absolute-overprecision measure used in the experiment. Normalizing by the sum of the two errors gives relative overprecision,

$$ROP_i = \frac{CAE_i - SE_i}{CAE_i + SE_i} = \frac{\sqrt{\omega_i} - 1}{\sqrt{\omega_i} + 1}. \tag{3}$$

Equivalently,

$$\omega_i = \left(\frac{1 + ROP_i}{1 - ROP_i}\right)^2. \tag{4}$$

Behavior is governed by perceived prior variance, whereas overprecision is the calibration wedge relating perceived uncertainty to calibrated accuracy. In the experiment, subjective error is reported directly, whereas actual error is realized over a finite set of heterogeneous pictures. The empirical absolute- and relative-overprecision measures are therefore noisy proxies for the population objects in equations (2)–(3), rather than direct observations of the structural parameter ω_i.

The agent receives an unbiased public signal

$$x = \theta + \epsilon, \qquad \epsilon \sim N(0, \sigma_\epsilon^2), \tag{5}$$

where signal noise is independent of the prior error. Following the convention in the rational-inattention literature, we model imperfect processing with an additional noise term ψ_i:

$$s_i = x + \psi_i, \qquad \psi_i \sim N(0, \sigma_{\psi,i}^2). \tag{6}$$

A lower value of $\sigma_{\psi,i}^2$ corresponds to more precise signal processing. We assume that ψ_i is independent of θ, ϵ, and the prior error.

Given the perceived prior variance, the posterior mean is a weighted average of the processed signal and the prior mean:

$$\widetilde{\mathbb{E}}_i[\theta \mid s_i] = \beta_i s_i + (1 - \beta_i)\mu_{\theta,i}, \tag{7}$$

where the signal weight is

$$\beta_i = \frac{\widetilde{\sigma}_{\theta,i}^2}{\widetilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2 + \sigma_{\psi,i}^2}. \tag{8}$$

We refer to β_i as the update rate. Strictly speaking, the posterior belief is a distribution; equation (7) is its expectation.

2.2 Endogenous attention

Agents determine the precision of signal processing, $\sigma_{\psi,i}^{-2}$, by trading off perceived posterior accuracy against attention costs. More formally, they solve

$$\max_{\sigma_{\psi,i}^{-2}} -\phi_i \widetilde{\mathrm{Var}}_i(\theta \mid s_i) - \widetilde{C}_i(\sigma_{\psi,i}^{-2}), \tag{9}$$

where $\phi_i > 0$ scales the incentive to hold an accurate posterior. Here, $\widetilde{\mathrm{Var}}_i(\theta \mid s_i)$ is the posterior variance under the agent's perceived distribution, and $\widetilde{C}_i$ is the perceived cost of processing the signal. The tildes indicate that both objects are evaluated using $\widetilde{\sigma}^2_{\theta,i}$ rather than $\sigma^2_{\theta,i}$. Because the posterior mean is the optimal point forecast under quadratic loss, the perceived conditional mean-squared error equals the perceived posterior variance.

Our benchmark model combines Gaussian beliefs, quadratic loss, and Shannon information costs. Attention costs are proportional to the entropy reduction generated by processing the signal. For a Gaussian variable, entropy is $H(z) = c + \frac{1}{2}\ln(\sigma_z^2)$, so the mutual information between the public signal and the processed signal implies the cost function

$$\widetilde{C}_i(\sigma_{\psi,i}^{-2}) = \frac{\lambda_i}{2} \ln\left(1 + \frac{\widetilde{\sigma}^2_{\theta,i} + \sigma^2_\epsilon}{\sigma^2_{\psi,i}}\right), \tag{10}$$

where $\lambda_i \geq 0$ is the marginal cost of attention. This specification gives transparent comparative statics. The mechanism does not require entropy costs to be taken literally; Appendix A considers a distinct class of increasing and convex processing costs and characterizes which benchmark comparative statics also hold in that environment.

Substituting (10) into (9) and expressing the choice in terms of the update rate gives the optimal update rate, with the zero-attention corner imposed,

$$\beta_i^* = \max\left\{0, \frac{\widetilde{\sigma}^2_{\theta,i}}{\widetilde{\sigma}^2_{\theta,i} + \sigma^2_\epsilon} - \frac{\lambda_i}{2\phi_i \widetilde{\sigma}^2_{\theta,i}}\right\}. \tag{11}$$

The first term is the Bayesian signal weight without additional processing noise. The second term is the attenuation generated by costly attention. Overprecision affects both terms because it lowers perceived prior variance.

2.3 Predictions

We first derive the comparative statics of the benchmark model. Assume that (11) has an interior solution. Substituting $\widetilde{\sigma}^2_{\theta,i} = \sigma^2_{\theta,i}/\omega_i$ yields three comparative statics. Unless stated otherwise, derivatives with respect to ω_i hold calibrated prior-error variance $\sigma^2_{\theta,i}$ and the other model primitives fixed. With the zero-attention corner imposed, Hypotheses 1 and 2 continue to hold as weak monotonicity results: an increase in overprecision or attention costs cannot raise the optimal update rate, although the corresponding marginal effect is zero while the corner remains binding. The derivative-based amplification result in Hypothesis 3 applies to interior choices; the cross-partial is not defined at the kink.

Hypothesis 1. Higher overprecision lowers the update rate:

$$\frac{\partial \beta_i^*}{\partial \omega_i} = -\frac{\sigma^2_\epsilon/\sigma^2_{\theta,i}}{(1 + \omega_i \sigma^2_\epsilon/\sigma^2_{\theta,i})^2} - \frac{\lambda_i}{2\phi_i \sigma^2_{\theta,i}} < 0. \tag{12}$$

An overprecise agent perceives less residual uncertainty in the prior and therefore weights the signal less.

Hypothesis 2. Higher attention costs lower the update rate:

$$\frac{\partial \beta_i^*}{\partial \lambda_i} = -\frac{\omega_i}{2\phi_i \sigma_{\theta,i}^2} < 0. \tag{13}$$

This is the standard rational-inattention force: as processing information becomes more expensive, the agent optimally processes less and updates less.

Equation (2) also gives $\widetilde{\sigma}_{\theta,i}^2 = (\pi/2)SE_i^2$. The interior cost effect can therefore be written as

$$\frac{\partial \beta_i^*}{\partial \lambda_i} = -\frac{1}{\pi \phi_i SE_i^2}.$$

Holding the incentive parameter fixed, agents who perceive their priors as more precise, and therefore report smaller subjective errors, are predicted to be more sensitive to an increase in processing costs.

Hypothesis 3. Overprecision amplifies the negative effect of attention costs:

$$\frac{\partial}{\partial \omega_i}\left(\frac{\partial \beta_i^*}{\partial \lambda_i}\right) = -\frac{1}{2\phi_i \sigma_{\theta,i}^2} < 0. \tag{14}$$

Within the interior region, this cross-partial is constant in ω_i and λ_i. Holding ϕ_i and $\sigma_{\theta,i}^2$ fixed, the benchmark therefore predicts the same marginal amplification throughout the interior region rather than an effect that varies with the initial levels of overprecision or attention costs. For two discrete cost levels $\lambda_H > \lambda_E$, if both choices are interior,

$$\beta_i^*(\omega_i, \lambda_H) - \beta_i^*(\omega_i, \lambda_E) = -\frac{(\lambda_H - \lambda_E)\omega_i}{2\phi_i \sigma_{\theta,i}^2}. \tag{15}$$

Thus, Feedback that reduces ω_i should make the Hard-Info penalty less negative, whereas Feedback that raises ω_i should make it more negative. This is the finite-difference counterpart of Hypothesis 3 and gives the predicted signs of the experimental interactions.

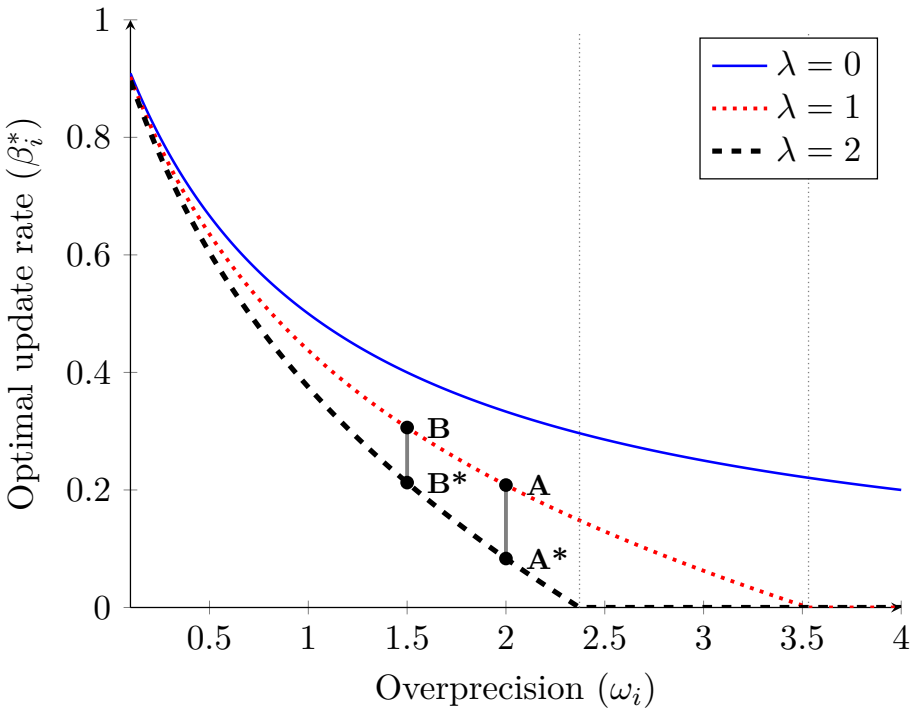
The common-interior qualification matters. Writing $a_i = \widetilde{\sigma}_{\theta,i}^2$, positive attention requires

$$\lambda_i < \overline{\lambda}(a_i) \equiv \frac{2\phi_i a_i^2}{a_i + \sigma_\epsilon^2}. \tag{16}$$

When Hard-Info is at the zero-attention corner but Easy-Info remains interior, the cost difference equals $-\beta_i^*(a_i, \lambda_E)$ and becomes less negative as a_i falls further. When both choices are at the corner, the difference is zero. The local cross-partial therefore does not imply that the finite Hard-Info effect grows monotonically with overprecision across the kink. Mapping Hypothesis 3 into the discrete experimental interaction additionally requires the relevant Easy-Info and Hard-Info potential choices to lie in the common interior region. This amplification effect is the benchmark model's distinctive interior prediction. Its robustness outside the benchmark also depends on how processing costs operate in the sequential environment and on the alternative cost technology considered below.

Figure 1 illustrates the comparative statics and the zero-attention corner. Within the common interior region, Hypothesis 3 corresponds to the widening vertical distance between the curves associated with different attention costs. Once the higher-cost choice reaches zero, that distance eventually narrows as the lower-cost choice also approaches the corner. The figure is stylized rather than a structural calibration of the experiment.

Figure 1: Optimal updating by varying overprecision and cost of attention



Notes: The figure plots the model-implied update rate for $\lambda_i \in \{0, 1, 2\}$, holding calibrated prior-error variance at $\sigma^2_{\theta,i} = 4$, signal noise at $\sigma^2_\epsilon = 4$, and the incentive parameter at $\phi_i = 2$. These values are illustrative rather than estimates from the experiment. The B–B^* segment shows the reduction in the update rate when λ_i rises from 1 to 2 at $\omega_i = 1.5$; the longer A–A^* segment shows the corresponding reduction at the higher overprecision level $\omega_i = 2$. Both comparisons are interior and illustrate Hypothesis 3. The vertical dotted lines mark the values of ω_i at which the positive-cost curves reach the zero-attention corner.

2.4 Model extensions

We derive the comparative statics above in the benchmark model. We use this environment because it yields a transparent expression for the update rate. Appendix A examines which of its comparative statics carry over to alternative environments.

First, the experimental task is naturally sequential: participants initially inspect a picture and later process a numerical signal. We consider a sequential extension in which the decision-maker chooses processing precision myopically at each stage, taking the current perceived prior variance as given. In this mapping, the benchmark ω_i is the calibration wedge in the prior held after viewing the picture and before processing the numerical signal; Appendix A.1 relates this post-picture wedge to calibration before picture processing. When Hard-Info changes only the cost of processing the second-stage numerical signal, as intended in the experiment, all three comparative statics carry over at an interior solution. We also consider a less experiment-specific case in which

a common cost parameter affects processing at both stages. In that case, Hypotheses 1 and 2 continue to hold, while Hypothesis 3 requires the sufficient restriction stated in Appendix A.1.

Second, we study an alternative class of increasing and convex processing costs. Let $q = \sigma_{\psi,i}^{-2}$ denote processing precision and let $C(q) = \lambda G(q)$, with $G'(q) > 0$ and $G''(q) \geq 0$. This class is distinct from, rather than a generalization that nests, the Shannon benchmark: expressed in terms of q, the Shannon cost in equation (10) depends on perceived prior variance and is concave in processing precision. We use convexity as a transparent reduced-form case in which attaining progressively greater processing precision requires increasing marginal resources. Convexity is sufficient rather than necessary for the first two comparative statics. In the alternative environment, optimal processing precision falls with λ and rises with perceived prior variance under standard regularity and interiority conditions. Hypotheses 1 and 2 therefore hold under this different cost technology as well. The cross-partial in Hypothesis 3 requires additional parameter and curvature restrictions. The amplification prediction is consequently a distinctive implication of the benchmark model rather than a mechanical implication of subjectively rational inattention.

3 Experimental design

3.1 Task and treatments

We test these predictions in an online experiment that independently randomizes two treatment dimensions: whether participants receive calibration feedback and the complexity of the information they subsequently receive. Participants estimate the average age of people in historical photographs. The images provide a natural basis for forming prior beliefs, while the continuous state allows us to elicit both a point estimate and an expected absolute error. Comparing expected and realized error then gives our empirical measure of overprecision, as in Bosch-Rosa et al. (2024).

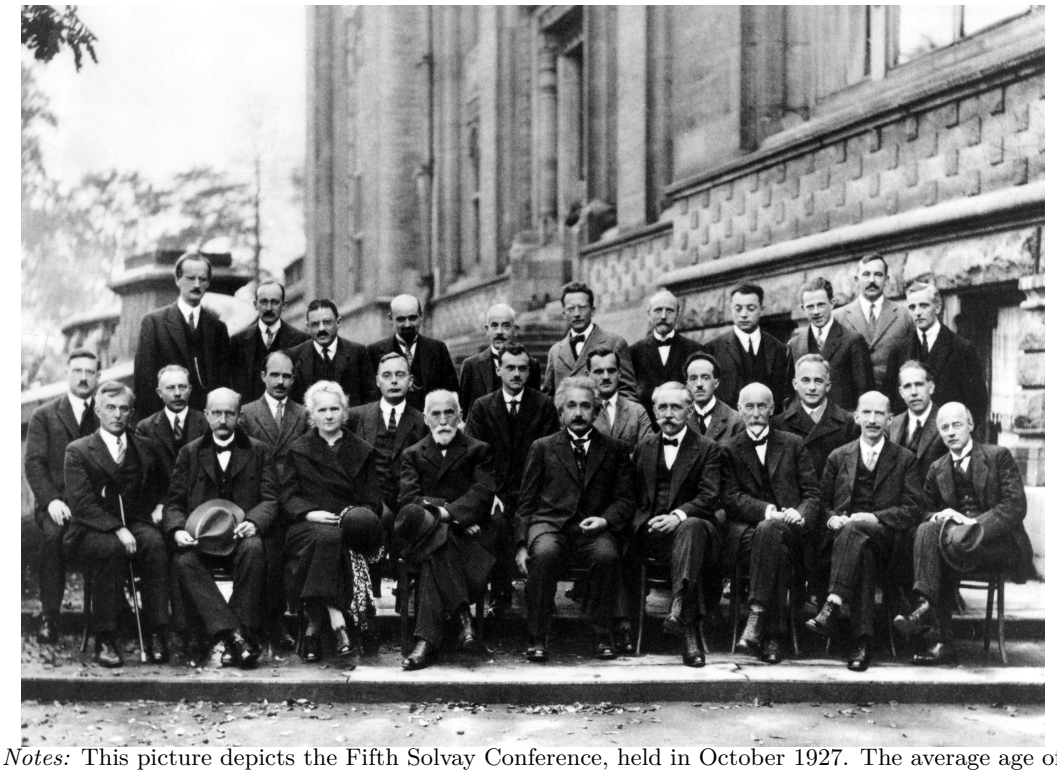
The experiment consists of two blocks. In Block 1, participants view five pictures, such as the one shown in Figure 2; Appendix B.1 reproduces the full set. For each picture k, participant i reports an estimate of the average age, denoted $prior_{i,k}$, and then reports the expected absolute distance between that estimate and the truth, denoted $se_{i,k}$. Both entries are restricted to the interval [0, 100]. The expected-error question is asked in the same units as the age estimate: participants state how far, in years, they think their answer is from the correct answer.[1] This elicitation avoids the well-known difficulties that arise when confidence intervals are elicited from heterogeneous populations (Teigen and Jørgensen, 2005; Moore et al., 2015).

At the end of Block 1, we randomly assign two-thirds of participants to receive calibration feedback.[2] The remaining third receive no feedback. Feedback is based on whether the participant's subjective errors were too small or too large in more than half of the first five

[1] The exact wording was: "*How far away from the correct answer do you think your answer is? Please state the distance in years.*"

[2] The feedback group is larger because the feedback message depends on Block 1 calibration; assigning two-thirds of participants to feedback provides adequate observations in both the Overprecise- and Underprecise-Feedback categories.

Figure 2: Example of a picture used in the online experiment



Notes: This picture depicts the Fifth Solvay Conference, held in October 1927. The average age of the people in the picture is 45.83 years. The youngest persons (Paul Dirac and Werner Heisenberg) are 25 years old, and the oldest (Hendrik Lorentz) is 74 years old.

pictures. The message reads:

> "In more than half of the last five pictures, your answer was [further away from the correct answer] [closer to the correct answer] than you expected."

We refer to the first message as *Overprecise Feedback* and the second as *Underprecise Feedback*. In both cases, the feedback is deliberately minimal and is intended to shift participants' perceived calibration without revealing correct answers or picture-specific errors. Providing more detailed feedback could teach participants about the task or change their beliefs about the distribution of ages.

In Block 2, participants complete five new picture tasks. As in Block 1, for each picture, they first report a prior estimate and expected absolute error. They then view a signal—a cloud-style display containing fifteen true ages of people in the picture—and submit a posterior estimate. We refer to this display as the signal throughout. Participants are randomly assigned to one of two information conditions. In Easy-Info, the signal contains only the fifteen ages. In Hard-Info, the same fifteen ages appear in the same positions, but the display also contains 85 irrelevant German nouns such as *Kunst*, *Brot*, or *Hemd* (see Figure 3 for an example). Thus, the signal has the same information content across conditions, but it is visually more complex in Hard-Info. Figure 4 summarizes the experimental design.

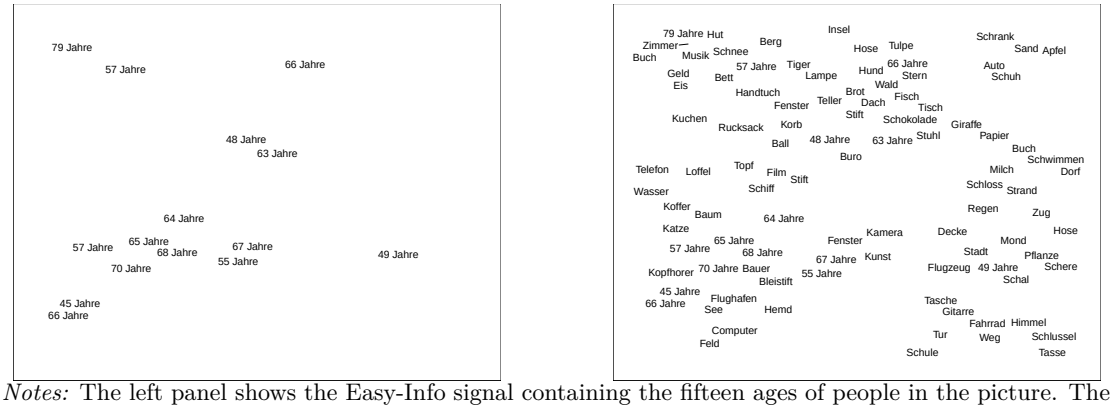
The design has three features that are particularly useful for linking the experiment to the model. First, expected absolute error directly measures perceived uncertainty in the same units as the belief. Because the age estimate and the expected error are both expressed in years, we can

compare perceived with realized error to measure overprecision, without relying on confidence intervals that many respondents find unfamiliar (Teigen and Jørgensen, 2005; Moore et al., 2015). Second, the decoy-word manipulation changes the difficulty of extracting information while leaving the signal itself unchanged. Differences across conditions therefore reflect responses to information complexity rather than differences in the signal's content or precision. Third, the continuous state distinguishes a belief's location from its perceived precision and generates variation in the distance between priors and signals. This makes it possible to study not only whether beliefs change, but how strongly participants respond to a given signal.

Figure 3: Example of the signal in the Easy-Info and Hard-Info treatments

(a) Easy-Info treatment

(b) Hard-Info treatment



Notes: The left panel shows the Easy-Info signal containing the fifteen ages of people in the picture. The right panel shows the Hard-Info signal, where the same ages appear in identical positions along with 85 random words. All decoy words are common German nouns (e.g., *Baum* = tree, *Buch* = book). All participants within each treatment saw the same set of randomly drawn ages and words.

3.2 Incentives and procedures

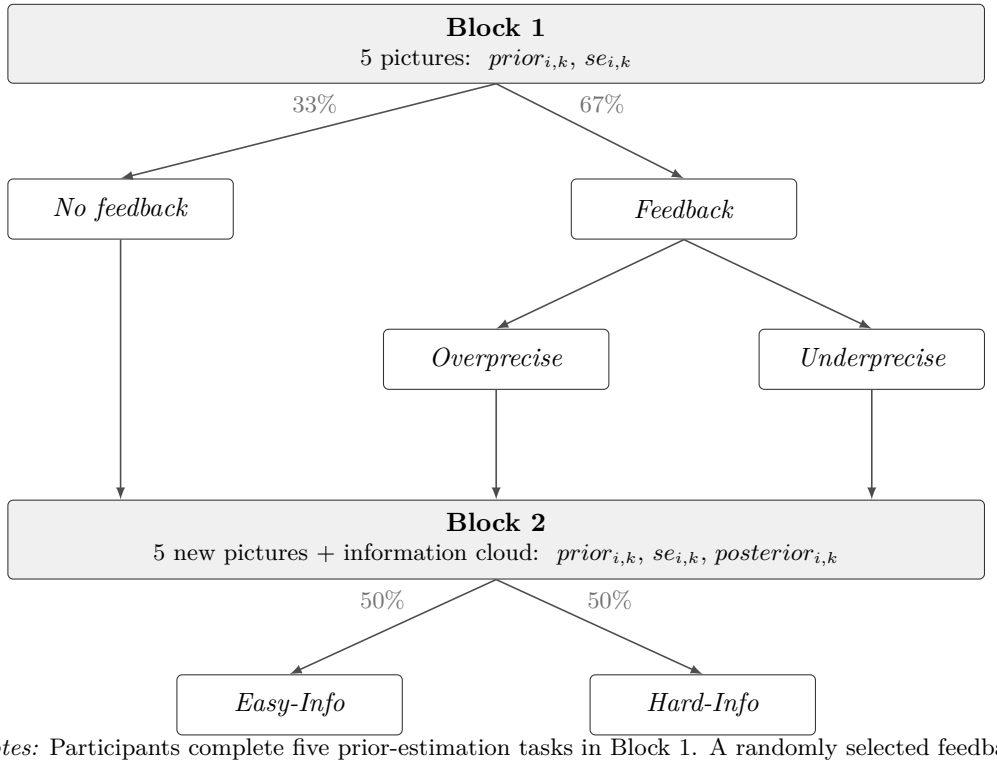
Bilendi fielded the survey using a representative sample of the German adult population. In addition to the standard survey payment, participants could earn a 10-euro accuracy bonus. We randomly selected 5% of participants to be eligible for the bonus and, if eligible, randomly drew one of their answers. The accuracy of that answer determined their probability of receiving the 10-euro bonus, implementing the binarized scoring rule (Hossain and Okui, 2013). Following Danz et al. (2022), participants were not required to read the formal scoring rule, but could access a detailed explanation via a link. Instead, the main instructions emphasized that more accurate answers increased the chance of receiving the bonus.[3]

3.3 Sample, exclusions, and pre-registration

We pre-registered the experimental design, hypotheses, target sample, exclusion rules, and main empirical analysis at AsPredicted #154320 before data collection. Before counting a survey as completed, we screened out participants who failed attention checks or answered fewer than two of four instruction-comprehension questions correctly. We collected 1,321 completed surveys

[3]In English translation: "At the end of the study, 5 percent of participants will be randomly selected for a bonus payment. If you are selected, one of your answers will be drawn at random. The closer this answer is to the correct answer, the higher your probability of receiving the 10-euro bonus."

Figure 4: Overview of the experimental design



Notes: Participants complete five prior-estimation tasks in Block 1. A randomly selected feedback group receives calibration feedback. In Block 2, participants complete five new tasks and update each estimate after seeing a signal. The information treatment randomizes signal visual complexity while holding its informational content fixed.

and then applied the pre-specified exclusion rules. We remove observations with infeasible subjective-error reports or age estimates outside the pre-specified 20–80 range, and exclude participants who violate either criterion more than once.[4] We also exclude speeders, defined as participants who completed more than half of the prior-estimation pages in less than 25 percent of the corresponding median completion time and in under 10 seconds.[5] After applying these rules, we retain the chronologically first 1,200 eligible completions, as pre-specified. The median participant spent 13 minutes completing the survey. Appendix Table C.1 reports sample demographics and population benchmarks.

The revised analysis differs from the pre-registered analysis in three main respects, none of which affects the substantive conclusions. Appendix D retains the corresponding pre-registered specifications and reports additional robustness checks.

- **Comparison groups.** The pre-analysis plan compared each Feedback arm with the pooled No-Feedback group. In the main text, we instead compare each Feedback arm with No-Feedback participants whose Block 1 responses would have triggered the same Feedback message. We use these comparison groups for both the Feedback manipulation checks and the belief-updating analysis. This avoids comparing a Feedback arm with a control group

[4] A subjective-error report is infeasible when it exceeds the largest possible absolute error given the reported prior and the 0–100 response scale, $\max\{prior_{i,k}, 100 - prior_{i,k}\}$.

[5] Response-time screening is commonly used to identify potentially low-engagement observations in web surveys. Greszki et al. (2015), for example, study very short completion times as an indicator of low-quality responding.

that combines participants with different initial calibration classifications. The corresponding pooled-control analyses are retained in Appendix D.

- **Feedback manipulation checks.** The pre-analysis plan used the change in average absolute overprecision from Block 1 to Block 2 as the outcome. In Section 4.2, we instead estimate Feedback effects on Block 2 outcomes conditional on the corresponding Block 1 measure, which accounts for baseline heterogeneity and improves precision. We also decompose absolute overprecision into subjective and actual error and report relative overprecision as an additional calibration measure. Appendix Table D.1 reproduces the original pre-registered manipulation-check analysis.
- **Inference.** The pre-analysis plan specified directional one-sided tests. We report two-sided tests throughout the main text for consistency and ease of interpretation. Appendix Table D.7 reports one-sided p-values for the pre-registered pooled-control specifications.

4 Results

4.1 Overview of the Data

Table 1 provides a first overview of participants' beliefs and updating across pictures. Average prior estimates are generally close to the true average ages, suggesting that participants engaged meaningfully with the task.[6] At the same time, participants tend to underestimate their own errors: in Block 1, reported subjective errors are generally smaller than realized errors, indicating overprecision on average. In Block 2, participants respond to the signal, with posterior beliefs moving toward it and ending closer to the truth than priors in all but one picture. Appendix Figures C.1–C.3 complement these averages by showing the full response distributions for each picture, including how posterior beliefs shift relative to priors after participants receive the signal.

4.2 Manipulation Checks

We begin by checking whether the two treatments affected the intended margins: participants' calibration in the Feedback treatment and signal processing difficulty in Hard-Info.

Feedback and overprecision

Following the Subjective Error Method of Bosch-Rosa et al. (2024), we define picture-level absolute overprecision as

$$op_{i,k} = \underbrace{|true_k - prior_{i,k}|}_{\text{realized absolute error}} - \underbrace{se_{i,k}}_{\text{expected absolute error}} . \tag{17}$$

[6] No prior report matched the full-precision average age except for Picture 3, the only picture with an integer average (54 years), where 30 of the 1,200 participants reported the correct value. No participant consistently reported the correct rounded answer across all ten pictures. Thus, while isolated exact answers occur, the responses do not show a pattern of consistently correct answers suggestive of systematic use of search engines or large language models.

Table 1: Summary statistics by picture: true ages, priors, errors, and posteriors

Block	Picture description	True age	Prior	Actual error	Subjective error	Posterior
1	Solvay Conference (1927)	45.83	56.09	10.99	5.23	–
			(8.62)	(7.44)	(3.83)	
1	Band Aid (1984)	27.89	29.84	4.86	3.99	–
			(5.91)	(4.11)	(3.07)	
1	Obama Cabinet (2009)	54.00	56.72	5.58	4.63	–
			(6.26)	(4.19)	(3.48)	
1	Paramount 75th Anniversary (1987)	52.32	54.10	7.45	5.25	–
			(8.76)	(5.23)	(4.01)	
1	Sino-German Consultations (2023)	57.04	53.07	5.93	4.53	–
			(6.17)	(4.97)	(3.09)	
2	G20 Summit (2017)	60.97	57.76	5.37	5.01	60.96
			(6.15)	(4.71)	(4.73)	(4.49)
2	The Irascibles (1950)	40.20	46.10	7.20	4.75	42.20
			(6.42)	(5.63)	(3.97)	(4.58)
2	Australian Labor Party (1901)	40.42	47.32	8.79	5.12	42.50
			(8.48)	(7.23)	(3.99)	(5.26)
2	Merkel Cabinet (2019)	52.05	51.47	4.81	4.86	51.65
			(5.99)	(4.49)	(4.81)	(4.42)
2	British Royal Family (2007)	48.65	49.32	7.66	5.91	49.95
			(9.47)	(5.97)	(4.66)	(8.30)

Notes: Means across up to $N = 1{,}200$ participants, with standard deviations in parentheses. All quantities are in years. The true average age is the mean of each person's age in completed years on the date the photograph was taken. Actual error is $|prior_{i,k} - \theta_k|$, where θ_k is the rounded task benchmark, differing from the true average by less than half a year. Subjective error is the reported expected absolute error. Priors and subjective errors precede the signal; posteriors follow it and pool Easy-Info and Hard-Info. Block 1 precedes Feedback and has no signal or posterior. Picture order is randomized within blocks.

The measure is positive when participants underestimate their error (overprecision) and negative when they overestimate it (underprecision).

Figure 5 plots mean picture-level absolute overprecision by picture order for the Overprecise, Underprecise, and No-Feedback groups. In Block 2, both Feedback groups move closer to calibration: overprecision declines among initially overprecise participants, while underprecision becomes less pronounced among initially underprecise participants. By contrast, the No-Feedback group remains relatively stable and, on average, overprecise. Appendix Table D.1 reports the pre-registered treatment estimates using the full No-Feedback group, while Appendix Figure D.1 plots picture-order-specific differences in absolute overprecision relative to the same control group.

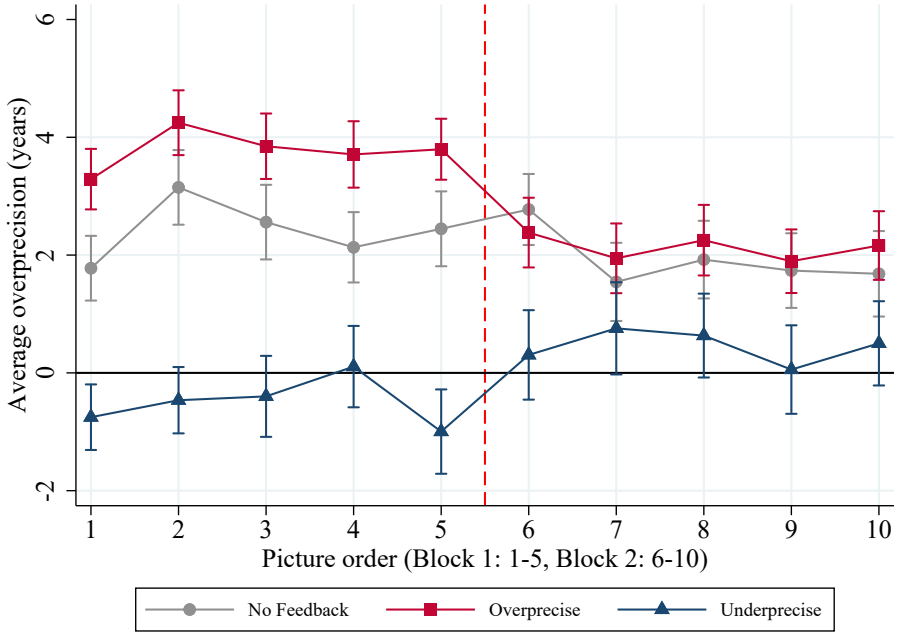
To estimate the effects of Feedback, we move from picture-level observations to respondent-level outcomes. Let $\mathcal{K}_{i,b}$ denote the set of pictures in block b for which respondent i's actual and subjective errors satisfy the registered validity restrictions, and let $n_{i,b} = |\mathcal{K}_{i,b}|$. We define block-level absolute overprecision as the average across these valid picture-level observations:

$$\overline{op}_{i,b} = \frac{1}{n_{i,b}} \sum_{k \in \mathcal{K}_{i,b}} op_{i,k} = \overline{ae}_{i,b} - \overline{se}_{i,b}, \tag{18}$$

where $\overline{ae}_{i,b}$ and $\overline{se}_{i,b}$ average respondent i's realized and expected absolute errors over the same set $\mathcal{K}_{i,b}$. We examine treatment effects on average subjective error, average actual error, and block-level absolute overprecision.

Because a given absolute discrepancy may represent different degrees of miscalibration

Figure 5: Raw average overprecision by Feedback group and picture order



Notes: For each Feedback group and picture order, the figure plots the sample mean of picture-level absolute overprecision ($op_{i,k}$) as defined in equation (17). The dashed line marks the start of Block 2.

depending on the magnitude of the underlying errors, we also consider relative overprecision:

$$rop_{i,b} = \frac{\overline{op}_{i,b}}{\overline{ae}_{i,b} + \overline{se}_{i,b}} = \frac{\overline{ae}_{i,b} - \overline{se}_{i,b}}{\overline{ae}_{i,b} + \overline{se}_{i,b}}. \tag{19}$$

Relative overprecision ranges from -1 to 1, with positive values indicating overprecision, negative values indicating underprecision, and zero indicating calibration.

For each outcome, we compare recipients of each Feedback message with No-Feedback participants whose Block 1 responses would have triggered the same feedback. We additionally control for the corresponding continuous Block 1 outcome to account for baseline heterogeneity and improve precision. We allow the relationship between the Block 1 and Block 2 outcomes to differ between initially overprecise and underprecise participants. Let F_i indicate assignment to Feedback and let R_i equal one if participant i's Block 1 responses trigger Overprecise rather than Underprecise Feedback. We define $Over_i = F_i R_i$ and $Under_i = F_i(1 - R_i)$, indicating receipt of the respective Feedback messages, while $Hard_i$ indicates assignment to Hard-Info. We use these treatment indicators throughout the paper. Specifically, we estimate

$$Y_{i,2} = \alpha + \beta_O Over_i + \beta_U Under_i + \gamma Y_{i,1} + \delta R_i + \eta\left(Y_{i,1} \times R_i\right) + \tau Hard_i + u_i. \tag{20}$$

Thus, β_O and β_U capture the Feedback effects within the initially overprecise and underprecise groups, respectively.

Table 2 shows a clear response among initially overprecise participants. Overprecise Feedback increases reported uncertainty while leaving realized error essentially unchanged. Accordingly,

Table 2: Feedback effects on subjective error, actual error, and overprecision

	Errors		Overprecision	
	Subjective	Actual	Absolute	Relative
Overprecise Feedback	0.508***	0.175	-0.339	-0.045**
	(0.147)	(0.215)	(0.263)	(0.019)
Underprecise Feedback	-0.276	-0.366	-0.046	-0.010
	(0.235)	(0.223)	(0.336)	(0.022)
Respondents	1,200	1,200	1,200	1,200
Symmetry test p-value	0.403	0.538	0.367	0.059
R^2	0.594	0.131	0.276	0.269

Notes: Outcomes are Block 2 averages, estimated by ANCOVA as in equation (20). Absolute and relative overprecision are defined in equations (18) and (19). The symmetry test evaluates $H_0 : \beta_{\text{Over}} + \beta_{\text{Under}} = 0$. Respondent-level estimates use robust standard errors, reported in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

the point estimates for both absolute and relative overprecision move toward calibration, although the relative estimate is more precise. By contrast, we find no clear response to Underprecise Feedback. The apparent asymmetry should be interpreted cautiously: formal tests provide only limited evidence that the two Feedback effects differ. This may partly reflect the larger Overprecise subgroup and the greater scope for adjustment among participants initially farther from calibration, as shown in Figure 5. Overall, Feedback increases reported uncertainty among initially overprecise participants, while the corresponding effect among initially underprecise participants is imprecisely estimated.

Information complexity and attention costs

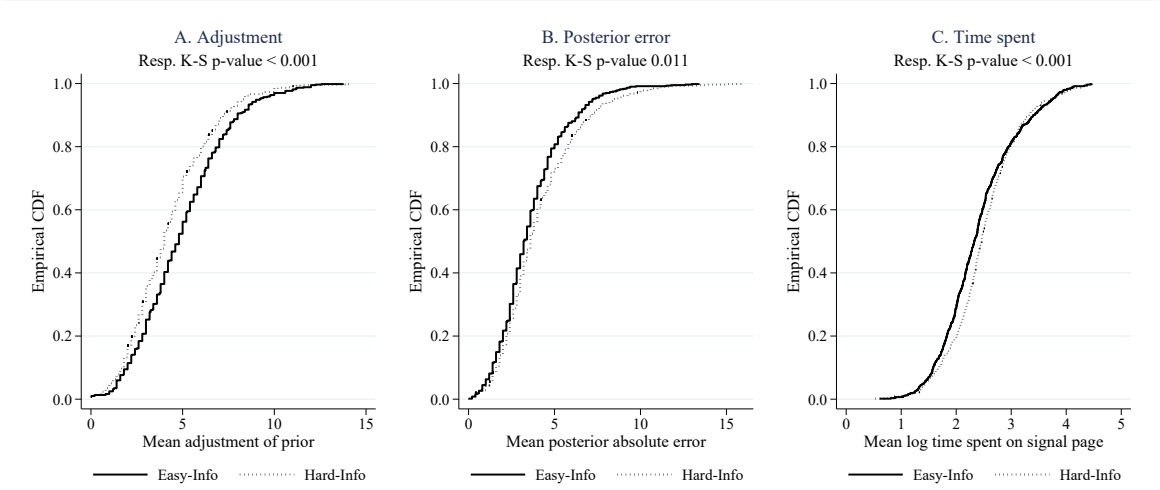
Figure 6 compares three outcomes across the two information conditions. Belief adjustment is the absolute revision after seeing the signal, $|posterior_{i,k} - prior_{i,k}|$; posterior error is the final absolute error, $|true_k - posterior_{i,k}|$; and processing time is the log time spent on the information page. Under Hard-Info, participants revise their beliefs less, end with less accurate posterior beliefs, and spend somewhat longer processing the information. Permutation Kolmogorov–Smirnov tests reject equality of the Easy-Info and Hard-Info distributions for all three outcomes.

Appendix Table E.3 provides additional evidence on how Hard-Info changes observed updating. Hard-Info increases the share of exact zero revisions by 3.67 percentage points, reduces absolute revisions by 0.735 years, and raises posterior absolute error by 0.491 years. Its effect on revisions that move beliefs away from the displayed signal is small and statistically insignificant. Lastly, responses under Hard-Info are not more dispersed around the estimated updating relationship. These results establish lower effective responsiveness, but they do not separately identify the underlying processing variance. In the model, higher processing noise also lowers the signal weight, so the residual variance of observed revisions, $\beta_i^2 \sigma_{\psi,i}^2$, need not increase monotonically with processing costs.

The evidence on processing time is mixed across specifications. Appendix Table E.1 shows that the OLS estimate of the Hard-Info effect is positive but imprecise, while the median and robust specifications indicate longer processing times. Appendix Table E.2 additionally

reports Feedback effects separately by information condition and initial calibration group; the Feedback-by-Hard-Info interactions are statistically insignificant. Longer response times can reflect greater effort, greater processing difficulty, or simply slower processing (Spiliopoulos and Ortmann, 2018). We therefore interpret processing time as complementary evidence on the decision process, rather than as a direct measure of effort or processing costs.

Figure 6: Information-complexity manipulation checks



Notes: Respondent-level distribution functions by information treatment. Picture-level values are trimmed at the pooled 1st and 99th percentiles before averaging by respondent. Subtitles report permutation Kolmogorov-Smirnov p-values.

4.3 Treatment Effects on Belief Updating

Having established how the treatments affect their intended margins, we next examine belief updating. We estimate

$$posterior_{i,k} - prior_{i,k} = \alpha + (signal_k - prior_{i,k})\Big(\beta_0 + \beta_1 Over_i + \beta_2 Under_i + \beta_3 Hard_i + \beta_4 Over_i \times Hard_i + \beta_5 Under_i \times Hard_i\Big) + \eta_k + u_{i,k}, \quad (21)$$

where $signal_k$ is the average age displayed in the information cloud for picture k, $Over_i$, $Under_i$, and $Hard_i$ are the treatment indicators defined above, and η_k are picture fixed effects.

The treatment interactions allow responsiveness to the displayed signal to vary across Feedback and information-complexity conditions. In particular, β_1 and β_2 capture how Feedback changes responsiveness under Easy-Info, β_3 captures the effect of Hard-Info among No-Feedback participants, and β_4 and β_5 test whether the effect of Hard-Info differs with Feedback.

To connect this specification to the model, recall that the posterior mean depends on the latent processed signal $s_i = x + \psi_i$, whereas the experiment records the objective displayed signal x. If the reported posterior equals the agent's perceived posterior mean, equation (7) implies

$$posterior_{i,k} - prior_{i,k} = \beta_{i,k}(signal_k - prior_{i,k}) + \beta_{i,k}\psi_{i,k}. \quad (22)$$

Under mean-zero processing noise that is independent of the displayed signal–prior gap, the conditional mean revision is $\beta_{i,k}(signal_k - prior_{i,k})$. The coefficients on the displayed signal–prior gap in equation (21) therefore measure effective signal responsiveness.[7] With heterogeneous update rates, these coefficients are linear-projection averages rather than every participant's structural $\beta_{i,k}$.

Table 3 reports the estimates. Column 1 compares Overprecise Feedback recipients with No-Feedback participants who were likewise classified as overprecise in Block 1, while column 2 provides the analogous comparison for Underprecise Feedback. Appendix Table D.3 reports the pre-registered specification using the pooled No-Feedback group.

Table 3: Belief updating by treatment assignment

	Initially overprecise	Initially underprecise
Signal − Prior	0.588***	0.615***
	(0.037)	(0.038)
Overprecise Feedback × (Signal − Prior)	0.075*	
	(0.044)	
Underprecise Feedback × (Signal − Prior)		-0.036
		(0.048)
Hard-Info × (Signal − Prior)	-0.124**	-0.168***
	(0.052)	(0.065)
Overprecise Feedback × Hard-Info × (Signal − Prior)	-0.026	
	(0.064)	
Underprecise Feedback × Hard-Info × (Signal − Prior)		0.109
		(0.079)
Constant	1.074***	1.205***
	(0.151)	(0.154)
Observations	3,590	2,356
Respondents	725	475
R^2	0.569	0.573

Notes: The outcome is posterior minus prior. Signal − Prior is the displayed signal average minus the prior. Each column compares Feedback with No-Feedback within the indicated Block 1 calibration group. Both models include picture fixed effects. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Under Easy-Info, Overprecise Feedback provides suggestive evidence of increased signal responsiveness relative to the matched No-Feedback group. We find no clear effect of Underprecise Feedback under Easy-Info. This pattern mirrors the manipulation checks in Section 4.2, where the Overprecise Feedback arm also produces a clearer change in reported uncertainty and calibration.

> *Result 1.* Under Easy-Info, we find suggestive evidence that Overprecise Feedback increases signal responsiveness, while no clear effect of Underprecise Feedback is detected.

Hard-Info clearly affects belief updating, reducing signal responsiveness, consistent with the model's prediction that greater processing costs reduce the weight placed on new information.

[7] We do not observe participants' perceived signal, so extraction, weighting, and effort are not separately identified. Interpreting responsiveness as a structural signal weight requires the assumptions stated above.

Result 2. Hard-Info reduces signal responsiveness relative to Easy-Info.

The interaction estimates do not support Hypothesis 3. Under the benchmark model's common-interior mapping to the discrete treatments, if Overprecise Feedback reduces overprecision, it should weaken Hard-Info's negative effect, implying a positive $Over_i \times Hard_i$ interaction. Conversely, Underprecise Feedback should strengthen the effect of Hard-Info, implying a negative $Under_i \times Hard_i$ interaction. Instead, the estimated interactions have the opposite signs in both matched comparisons and are statistically indistinguishable from zero. We therefore find no evidence that Feedback changes the effect of information complexity on signal responsiveness in the direction predicted by the model.

Result 3. We find no statistically significant effect of feedback on sensitivity to information complexity.

Taken together, the results show that Hard-Info reduces signal responsiveness and provide suggestive evidence that Overprecise Feedback increases it. They do not, however, establish the predicted amplification effect. The pre-registered and alternative specifications in Appendix D yield a similar qualitative pattern.[8]

5 Exploratory Analyses

To investigate the null interaction in Section 4.3, we examine possible statistical, theoretical, and design-based explanations through exploratory analyses. We also briefly summarize a complementary analysis of the predicted relationship using measured overprecision, which avoids reliance on the Feedback first stage but yields correlational estimates.

Statistical explanations. One possibility is that the experiment has limited power to detect the interaction predicted by Hypothesis 3. Testing whether the effect of information complexity differs across Feedback conditions is typically more demanding than detecting individual treatment effects, so a factorial experiment may detect those effects while remaining underpowered for their interaction (List et al., 2011; Muralidharan et al., 2025). In Appendix F.1, we assess the power of the pre-registered pooled-control interaction test using simulations based on our experimental design. The simulations indicate limited power to detect smaller interactions. Although the estimated interactions have signs opposite to those predicted, their confidence intervals include small effects in the predicted directions. Sampling variation could therefore account for the opposite signs, but the estimates provide no affirmative support for the amplification prediction.

A related concern is measurement error in reported beliefs. Measurement error in experimentally elicited characteristics can attenuate estimated relationships and bias correlations (Gillen et al., 2019), while belief-elicitation procedures can generate systematic reporting distortions

[8] These include changes to comparison groups and covariates (Table D.3), picture-specific estimates (Table D.4), an alternative updating equation (Table D.5), and a specification allowing treatment-specific intercepts and additional slope heterogeneity (Table D.6).

(Danz et al., 2022). In particular, Bland and Rosokha (2024) show that rounding and heaping can materially affect estimated updating parameters. In our setting, the reported prior enters both belief revision and the signal–prior gap, so reporting noise can distort the estimated updating relationship. To examine treatment interactions without using the reported prior on both sides of the regression, Appendix Table F.2 considers specifications that omit the signal–prior gap and its interactions. The first retains belief revision as the outcome, while the second and third use posterior absolute error, with the third additionally controlling for respondents' average Block 1 actual error. The treatment interactions are statistically insignificant throughout. These specifications provide complementary evidence on revisions and accuracy, although they do not correct reporting error in the original signal-responsiveness regression.

Theoretical explanations. Another possibility is that observed belief revisions do not correspond exactly to the updating process described by the model. Of the 5,946 picture-level revisions, 18.9 percent leave the prior unchanged and 12.7 percent move away from the displayed signal. Zero revisions may reflect either the model's zero-attention corner or coarse reporting. Wrong-way revisions may arise from processing noise within the model or from departures from Bayesian updating (Ortoleva, 2024).[9] Neither pattern alone therefore establishes a departure from the benchmark model. To assess whether the interaction estimates are sensitive to these patterns, we re-estimate the treatment specification first excluding exact zero revisions and then excluding both zero and wrong-way revisions (Appendix Table F.3). Neither restriction provides support for the predicted interaction.

More generally, participants may update conservatively, placing less weight on their perceived signal than Bayesian updating would prescribe (Benjamin, 2019), or information complexity may generate systematically biased internal representations of the signal. Experimental evidence shows that complexity and cognitive uncertainty can attenuate responses to objective information and generate departures from benchmark Bayesian updating (Charness and Levin, 2005; Enke and Graeber, 2023). These mechanisms could contribute to muted or irregular revisions, but the available evidence does not establish whether they account for the insignificant interaction.

The null interaction could also reflect departures from the benchmark assumptions about processing costs. Rational-inattention theory allows cost structures beyond the Shannon specification (Caplin et al., 2022), and experimental evidence suggests that behavior can depart from predictions based on costs that are linear in Shannon entropy (Dean and Neligh, 2023). Our extensions, discussed in Section 2.4 and derived in Appendix A, preserve Hypotheses 1 and 2 under the maintained conditions. Hypothesis 3 also holds at an interior solution in the extension that most closely matches our experiment, where information complexity changes only the cost of processing the second-stage signal. When a common cost parameter affects both processing stages, however, or when processing costs follow the alternative convex specification, amplification requires additional restrictions. The two direct effects can therefore hold even when the predicted amplification does not.

[9] Appendix F.2 derives the model-implied probability of wrong-way revisions.

Design-based explanations. A third possibility is that the experimental treatments affect more than the model primitives they are intended to manipulate. Feedback is designed to change perceived prior precision, while Hard-Info is designed to change processing costs. The interaction maps onto the model's precision-by-cost effect only if the treatments operate through these intended channels. If either intervention also changes other determinants of updating, the estimated interaction may capture these additional effects as well.[10]

Performance feedback may affect motivation, effort, and subsequent information processing through changes in self-confidence, beyond its effect on perceived prior precision (Bénabou and Tirole, 2002; Compte and Postlewaite, 2004; Möbius et al., 2022). Appendix F.3 examines one possible spillover: Feedback may also change perceived processing costs. If Overprecise Feedback makes the signal seem harder to process, participants may respond less to it, working against the increase in responsiveness predicted by lower perceived prior precision. In the appendix's local approximation, the effect on the Feedback-by-Hard-Info interaction depends on how the cost spillover differs between Hard-Info and Easy-Info. An overall increase in perceived processing costs therefore does not by itself establish that the interaction understates amplification.

Appendix Table F.4 examines whether Feedback affects reported task difficulty. After the experiment, participants rated how difficult they found viewing the pictures and information clouds on a scale from 1 to 5. Overprecise Feedback is associated with greater reported difficulty, but the Feedback-by-Hard-Info interactions are imprecisely estimated. Because the question combines picture viewing and signal processing, it does not isolate the perceived cost of processing the numerical signal or directly identify the model's processing-cost parameter. The findings therefore do not establish that Feedback changes processing costs differently under Hard-Info and Easy-Info or that such a difference explains the null interaction.

A related design limitation concerns the Feedback first stage, particularly for the Underprecise Feedback arm. Section 4.2 documents a clearer response to Overprecise Feedback, while the response to Underprecise Feedback is estimated less precisely. Responses may differ across participants, and a message based on only five pictures may provide a noisy indication of their broader calibration. Related work documents asymmetric processing of favorable and unfavorable feedback about personal attributes (Eil and Rao, 2011; Möbius et al., 2022). Our calibration-feedback design differs from those settings, and our formal tests provide limited evidence of asymmetric responses. If Underprecise Feedback changes perceived prior precision only slightly, the predicted interaction may be difficult to detect.

Complementary analysis. Given these limitations of the Feedback-based interaction test, we also examine whether the effect of information complexity on signal responsiveness varies with measured overprecision. This approach avoids relying on Feedback-induced changes in calibration. Appendix F.4 reports the specifications, estimates, and robustness checks. The estimated interactions with Hard-Info are negative using both pre-Feedback Block 1 overprecision and Block 2 overprecision. Restricting the Block 2 specification to No-Feedback participants also yields a negative, imprecisely estimated interaction. These patterns are directionally consistent

[10] This concern is well known in information-provision experiments, where treatments can affect behavior through channels beyond the targeted belief (Haaland et al., 2023).

with the amplification prediction, but statistical significance is not robust across specifications. Moreover, because overprecision is measured rather than randomly assigned, the estimates do not identify a causal effect of overprecision.

6 Conclusion

This paper studies how overprecision shapes the value of information. In the model, agents choose how much attention to allocate based on how precise they believe their prior to be. Overprecision makes the prior appear too precise, reducing the perceived value of new information and potentially making updating more sensitive to processing costs. The experiment separately randomizes two interventions targeting these margins, using Feedback to shift perceived prior precision and information complexity to vary processing costs, and tests whether they interact as predicted by the benchmark model.

The experimental results are partly consistent with the model's predictions. Among initially overprecise participants, calibration feedback increases reported uncertainty and provides suggestive evidence of greater signal responsiveness, while no comparably clear response emerges among initially underprecise participants. Increasing information complexity reduces signal responsiveness and worsens posterior accuracy, but the interactions with Feedback are statistically insignificant. To interpret this result, we conduct exploratory theoretical and empirical analyses. The model extensions show that amplification is more sensitive to assumptions about processing costs than the two direct predictions, while the empirical diagnostics examine design and measurement features that may make the interaction harder to detect or interpret. Complementary estimates using measured overprecision yield negative interactions with information complexity, but these estimates are correlational and not robustly statistically significant. These analyses therefore help interpret the treatment results without establishing the predicted amplification effect.

The model highlights a metacognitive source of inattention. Overprecise agents may optimize given their beliefs yet undervalue new information and allocate too little attention relative to a calibrated Bayesian benchmark. Future experiments could use richer feedback to shift perceived precision more strongly and separate elicitation conditions to distinguish signal extraction from weighting. Independently varying accuracy incentives and perceived precision could also test the model's prediction that, at an interior solution, overprecision increases responsiveness to incentives. These mechanisms matter for information design. Simplifying information can encourage updating. In the model, higher accuracy incentives can mitigate attenuation from processing costs, but they do not correct misperceived prior precision or eliminate its direct effect on signal weighting. Changing how people assess the precision of their knowledge may therefore be as important as making information easier to process.

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A Model extensions

This appendix provides the two robustness extensions discussed in Section 2.4. The first studies how the static comparative statics carry over to a sequential, multi-signal environment. The second considers a distinct class of increasing and convex processing costs. The latter is an alternative cost technology rather than a class that nests the Shannon benchmark.

A.1 Multi-signal environment

We consider a myopic sequential decision-maker who chooses processing precision at each stage, takes the current perceived prior variance as given, and does not internalize how current attention affects the value of future information. This myopia assumption makes the stage problem recursive; a forward-looking decision-maker could instead adjust early attention in anticipation of later processing costs.

Suppose there are $k = 1, 2, \ldots$ signals processed sequentially and conditionally independently given θ. Before processing signal k, the decision-maker chooses the precision of idiosyncratic processing noise, $\sigma_{\psi,i}^{-2}(k)$, and perceives

$$s_i(k) = \theta + \epsilon(k) + \psi_i(k), \qquad \epsilon(k) \sim N(0, \sigma_\epsilon^2(k)), \quad \psi_i(k) \sim N(0, \sigma_{\psi,i}^2(k)),$$

where the errors are mutually independent across stages and independent of θ. Allowing $\lambda_i(k)$ and $\phi_i(k)$ to vary by stage, the static solution applies recursively:

$$\beta_i^*(k) = \max\left\{0,\ \frac{\widetilde{\sigma}^2_{\theta,i|s(k-1)}}{\widetilde{\sigma}^2_{\theta,i|s(k-1)} + \sigma_\epsilon^2(k)} - \frac{\lambda_i(k)}{2\phi_i(k)\,\widetilde{\sigma}^2_{\theta,i|s(k-1)}}\right\}, \tag{23}$$

$$\widetilde{\sigma}^2_{\theta,i|s(k)} = (1 - \beta_i^*(k))\widetilde{\sigma}^2_{\theta,i|s(k-1)}, \qquad \widetilde{\sigma}^2_{\theta,i|s(0)} = \frac{\sigma^2_{\theta,i}}{\omega_i}. \tag{24}$$

To show how the static model extends to the sequential setting, consider the two-signal case. In the experimental mapping, the first stage represents processing the picture to form a prior and the second represents processing the numerical signal about that same picture. The stages do not represent the five Block 2 pictures, which concern different underlying states. It is important to distinguish calibration before and after picture processing: the experiment elicits the prior only after the first stage. From here on, suppose that (23) has an interior solution. To keep the algebra readable, let $v_\theta \equiv \sigma^2_{\theta,i}$, $v_{\epsilon k} \equiv \sigma_\epsilon^2(k)$, and $\omega \equiv \omega_{i,0}$, where $\omega_{i,0}$ is the calibration wedge before the picture is processed.

We first consider the experiment-aligned case in which the cost and incentive parameters may differ across stages. Write these parameters as (λ_1, ϕ_1) and (λ_2, ϕ_2). At an interior first-stage solution,

$$\beta^*(1) = \frac{v_\theta}{v_\theta + \omega v_{\epsilon 1}} - \frac{\lambda_1 \omega}{2\phi_1 v_\theta}.$$

The perceived variance entering stage 2 is therefore

$$T(\omega, \lambda_1) \equiv (1 - \beta^*(1))\frac{v_\theta}{\omega} = \frac{v_\theta v_{\epsilon 1}}{v_\theta + \omega v_{\epsilon 1}} + \frac{\lambda_1}{2\phi_1}. \tag{25}$$

To connect this object to the post-picture prior elicited in the experiment, let $r_1 = v_{\epsilon 1} + \sigma^{2*}_{\psi,i}(1)$ denote total effective first-stage signal noise. At an interior solution, $\beta^*(1) = (v_\theta/\omega)/(v_\theta/\omega + r_1)$. The calibrated variance of the error in the post-picture prior is therefore

$$V_1 = \big(1 - \beta^*(1)\big)^2 v_\theta + \beta^*(1)^2 r_1.$$

Defining the post-picture calibration wedge by $\omega_{i,1} = V_1/T$ gives

$$\omega_{i,1} = \omega_{i,0}\big(1 - \beta^*(1)\big) + \beta^*(1). \tag{26}$$

Thus, first-stage updating generally moves the calibration wedge toward one, and the pre-picture and post-picture wedges are not the same object.

At an interior second-stage solution, the weight on the numerical signal is

$$\beta^*(2) = \frac{T}{T + v_{\epsilon 2}} - \frac{\lambda_2}{2\phi_2 T}. \tag{27}$$

The direct model-to-experiment mapping takes $(V_1, \omega_{i,1})$ as the prior state when the numerical signal arrives, so that $T = V_1/\omega_{i,1}$. Substituting this expression into equation (27) and holding calibrated post-picture variance V_1 fixed gives

$$\frac{\partial \beta^*(2)}{\partial \omega_{i,1}} = -\frac{V_1 v_{\epsilon 2}}{(V_1 + \omega_{i,1} v_{\epsilon 2})^2} - \frac{\lambda_2}{2\phi_2 V_1} < 0,$$

$$\frac{\partial \beta^*(2)}{\partial \lambda_2} = -\frac{\omega_{i,1}}{2\phi_2 V_1} < 0, \qquad \frac{\partial}{\partial \omega_{i,1}}\left(\frac{\partial \beta^*(2)}{\partial \lambda_2}\right) = -\frac{1}{2\phi_2 V_1} < 0.$$

These are the three stage-two comparative statics in terms of the post-picture calibration wedge targeted by the experiment's calibration measures.

For completeness, pre-picture calibration propagates into the second-stage choice through

$$T_\omega = -\frac{v_\theta v_{\epsilon 1}^2}{(v_\theta + \omega v_{\epsilon 1})^2} < 0.$$

The Hard-Info treatment changes the difficulty of processing the numerical cloud after the prior has been reported. Its closest counterpart is therefore an increase in λ_2 holding λ_1 fixed. Equation (27) then implies $\partial \beta^*(2)/\partial \omega < 0$, $\partial \beta^*(2)/\partial \lambda_2 < 0$, and $\partial^2 \beta^*(2)/(\partial \omega\, \partial \lambda_2) < 0$. Thus, under interiority and the maintained myopia assumption, all three benchmark hypotheses also carry through from pre-picture calibration when information complexity changes only the second-stage processing cost. Equation (26) clarifies, however, that these derivatives with respect to the initial wedge are distinct from the post-picture calibration object targeted by the experimental measures. The direct experimental counterpart is the stage-two result stated above in terms of $\omega_{i,1}$.

We next consider a common-cost robustness case in which $\lambda_1 = \lambda_2 = \lambda$ and $\phi_1 = \phi_2 = \phi$. This comparative static is less closely aligned with the treatment because it allows information complexity to reduce processing of the picture before the numerical cloud appears. It is nevertheless useful for studying substitution of attention across stages. The derivations below

show that Hypotheses 1 and 2 continue to hold under interiority, while Hypothesis 3 requires an additional condition.

Hypothesis 1 (multi-signal). *Increasing overprecision leads to a lower update rate after the second signal.*

Support. In the common-cost case, equations (25) and (27) become

$$T(\omega,\lambda)=\frac{v_\theta v_{\epsilon 1}}{v_\theta+\omega v_{\epsilon 1}}+\frac{\lambda}{2\phi},\qquad \beta^*(2)=B(T,\lambda):=\frac{T}{T+v_{\epsilon 2}}-\frac{\lambda}{2\phi T}. \tag{28}$$

The relevant derivatives are

$$T_\omega=-\frac{v_\theta v_{\epsilon 1}^2}{(v_\theta+\omega v_{\epsilon 1})^2}<0,\qquad T_\lambda=\frac{1}{2\phi},\qquad B_T=\frac{v_{\epsilon 2}}{(T+v_{\epsilon 2})^2}+\frac{\lambda}{2\phi T^2}>0.$$

It follows immediately that

$$\frac{\partial\beta^*(2)}{\partial\omega}=B_T T_\omega<0,$$

which is qualitatively identical to the static prediction.

Hypothesis 2 (multi-signal). *Increasing attention costs lead to a lower update rate after the second signal.*

Support. The total effect combines the positive indirect channel through T_λ with the negative direct cost effect. Differentiating equation (28) gives

$$\frac{\partial\beta^*(2)}{\partial\lambda}=\frac{1}{2\phi}\left[\frac{v_{\epsilon 2}}{(T+v_{\epsilon 2})^2}-\frac{1}{T}+\frac{\lambda}{2\phi}\frac{1}{T^2}\right]. \tag{29}$$

Interiority in the second stage, $\beta^*(2)>0$, implies

$$\frac{\lambda}{2\phi}<\frac{T^2}{T+v_{\epsilon 2}},$$

and hence

$$\frac{\lambda}{2\phi T^2}<\frac{1}{T+v_{\epsilon 2}}.$$

Applying this inequality to the bracketed term in equation (29) gives

$$\begin{aligned}\frac{v_{\epsilon 2}}{(T+v_{\epsilon 2})^2}-\frac{1}{T}+\frac{\lambda}{2\phi T^2}&<\frac{v_{\epsilon 2}}{(T+v_{\epsilon 2})^2}-\frac{1}{T}+\frac{1}{T+v_{\epsilon 2}}\\&=-\frac{v_{\epsilon 2}^2}{T(T+v_{\epsilon 2})^2}<0.\end{aligned}$$

Therefore, $\partial\beta^*(2)/\partial\lambda<0$ at every interior second-stage solution. The indirect effect of a higher common attention cost—less first-stage processing leaves greater perceived variance for stage 2—partly offsets the direct second-stage cost effect, but cannot reverse it. Hypothesis 2 thus carries over under interiority alone.

Hypothesis 3 (multi-signal). *Overprecision amplifies the negative effect of attention costs on the update rate.*

Support. This hypothesis concerns the cross-derivative

$$\frac{\partial}{\partial \omega}\left(\frac{\partial \beta^*(2)}{\partial \lambda}\right).$$

Using equation (29), we can write

$$\frac{\partial}{\partial \omega}\left(\frac{\partial \beta^*(2)}{\partial \lambda}\right) = \frac{1}{2\phi}\frac{\partial}{\partial T}\underbrace{\left[\frac{v_{\epsilon 2}}{(T+v_{\epsilon 2})^2} - \frac{1}{T} + \frac{\lambda}{2\phi}\frac{1}{T^2}\right]}_{=:\ \mathcal{H}(T,\lambda)} \cdot \frac{\partial T}{\partial \omega}. \tag{30}$$

From above, $\partial T/\partial \omega < 0$. The sign is therefore governed by the partial derivative of $\mathcal{H}(T,\lambda)$ with respect to T:

$$\frac{\partial \mathcal{H}(T,\lambda)}{\partial T} = -\frac{2v_{\epsilon 2}}{(T+v_{\epsilon 2})^3} + \frac{1}{T^2} - \frac{\lambda}{\phi}\frac{1}{T^3}.$$

Let $u := T/(T+v_{\epsilon 2})$ and $r := \lambda/(\phi T)$. Then

$$\frac{2v_{\epsilon 2}}{(T+v_{\epsilon 2})^3} = \frac{2(1-u)u^2}{T^2},$$

so

$$\frac{\partial \mathcal{H}}{\partial T} = \frac{1}{T^2}\left[1 - 2(1-u)u^2 - r\right].$$

Because $\partial T/\partial \omega < 0$, equation (30) implies the exact sign condition

$$\frac{\partial}{\partial \omega}\left(\frac{\partial \beta^*(2)}{\partial \lambda}\right) < 0 \quad \Longleftrightarrow \quad r < 1 - 2(1-u)u^2. \tag{31}$$

Moreover, $\beta^*(2) = u - r/2$. A reversal of Hypothesis 3 requires $r > 1 - 2(1-u)u^2$ and therefore

$$\beta^*(2) < u - \frac{1}{2} + (1-u)u^2 < \frac{1}{2},$$

where the final inequality follows because $(1-u)(1-u^2) > 0$ for $u \in (0,1)$. Thus, an interior update rate below one-half is necessary, though not sufficient, for the common-cost extension to reverse the amplification effect.

The function $m(u) := 2(1-u)u^2$ attains its maximum $8/27$ at $u = 2/3$, so the right-hand side of equation (31) is bounded below by $19/27$. Hence

$$\frac{\partial \mathcal{H}}{\partial T} \geq \frac{1}{T^2}\left(\frac{19}{27} - \frac{\lambda}{\phi T}\right).$$

Thus, $\lambda/(\phi T) < 19/27$ is a parameter-free sufficient condition. This condition is equivalent to a bound on the primitive cost–incentive ratio. Since

$$T = \frac{v_\theta v_{\epsilon 1}}{v_\theta + \omega v_{\epsilon 1}} + \frac{\lambda}{2\phi},$$

substitution and rearrangement give

$$\text{if} \quad \frac{\lambda}{\phi} < \frac{38}{35}\frac{v_\theta v_{\epsilon 1}}{v_\theta + \omega v_{\epsilon 1}}, \quad \text{then} \quad \frac{\partial}{\partial \omega}\left(\frac{\partial \beta^*(2)}{\partial \lambda}\right) < 0.$$

Under this condition, overprecision amplifies the negative effect of attention costs on the second-signal update rate. When the bound fails, the sign is determined by the sharper condition in equation (31); failure of the sufficient bound alone does not imply a reversal.

A.2 Alternative convex processing costs

This section examines whether the benchmark comparative statics also arise under a different processing-cost technology. We replace the Shannon cost with a class of increasing and convex costs and show that Hypotheses 1 and 2 continue to hold at an interior optimum. Convexity alone, however, is not sufficient to sign the cross-partial underlying Hypothesis 3. We therefore characterize additional sufficient conditions under which the benchmark amplification effect is preserved.

Suppressing individual subscripts, let $q \equiv \sigma_{\psi,i}^{-2}$ denote processing precision, $a \equiv \widetilde{\sigma}_{\theta,i}^2 = \sigma_{\theta,i}^2/\omega_i$ perceived prior variance, and $v_\epsilon \equiv \sigma_\epsilon^2$. The decision-maker solves

$$\max_{q \geq 0} \left\{ -\phi \, \widetilde{\mathrm{Var}}_i(\theta \mid s_i) - \lambda G(q) \right\}, \qquad G'(q) > 0, \quad G''(q) \geq 0.$$

We assume $\lambda > 0$, normalize $G(0) = 0$, and take G to be continuous on $[0, \infty)$ and twice continuously differentiable on $(0, \infty)$. The increasing convex cost is coercive, while the accuracy benefit is bounded, so an optimum exists. At $q = 0$, posterior variance and the signal weight are defined by their limits, a and zero, respectively. If the right derivative $G'(0+)$ is finite, the optimum is at the zero-processing corner when

$$\phi a^2 \leq \lambda G'(0+),$$

and is interior when the inequality is reversed. The derivative results below condition on an interior optimum $q^* > 0$. This specification does not nest the benchmark Shannon cost. Written in terms of q and a, equation (10) is

$$C^S(q, a) = \frac{\lambda}{2} \ln(1 + q(a + v_\epsilon)),$$

which depends on perceived prior variance and is concave in q. We use convexity to represent a reduced-form technology in which the marginal resource cost of processing precision is increasing. The exercise below asks whether the predictions survive in this distinct environment; it does not treat convexity as a property of the Shannon benchmark.

Under normal-normal updating, define

$$\widetilde{\mathrm{Var}}_i(\theta \mid s_i) = \frac{a(v_\epsilon + q^{-1})}{a + v_\epsilon + q^{-1}}, \qquad D(q, a) := 1 + q(a + v_\epsilon), \qquad H(q, a) := \phi \frac{a^2}{D(q, a)^2}.$$

The first-order condition for an interior optimum is

$$F(q, a, \lambda) \equiv \lambda G'(q) - H(q, a) = 0. \tag{32}$$

The relevant derivatives are

$$H_a = \frac{2\phi a(1 + qv_\epsilon)}{D(q, a)^3} > 0, \qquad H_q = -\frac{2\phi a^2(a + v_\epsilon)}{D(q, a)^3} < 0,$$
$$F_q = \lambda G''(q) - H_q(q, a) > 0.$$

Convexity is sufficient but not necessary for this inequality. The derivations of Hypotheses 1 and 2 below require only $G'(q) > 0$ and $F_q > 0$. They therefore also apply locally to nonconvex processing costs whenever the curvature of G is not sufficiently negative to violate this second-order condition. Applying the implicit function theorem gives

$$q^*_\lambda = -\frac{G'(q^*)}{F_q} < 0, \qquad q^*_a = \frac{H_a(q^*, a)}{F_q} > 0.$$

Thus, a higher marginal cost of attention reduces optimal processing precision, while greater perceived prior variance increases it.

The weight placed on the perceived signal and its relevant partial derivatives are

$$\beta(q, a) = \frac{a}{a + v_\epsilon + q^{-1}} = \frac{aq}{D(q, a)}, \qquad \beta_q = \frac{a}{D(q, a)^2} > 0, \qquad \beta_a = \frac{q(1 + qv_\epsilon)}{D(q, a)^2} > 0.$$

Consequently,

$$\frac{\partial \beta^*}{\partial \lambda} = \beta_q q^*_\lambda < 0, \qquad \frac{d\beta^*}{da} = \beta_a + \beta_q q^*_a > 0.$$

The first inequality establishes Hypothesis 2. For Hypothesis 1, because

$$a = \frac{\sigma_\theta^2}{\omega} \qquad \Longrightarrow \qquad \frac{\partial a}{\partial \omega} = -\frac{\sigma_\theta^2}{\omega^2} < 0,$$

we have

$$\frac{\partial \beta^*}{\partial \omega} = \frac{d\beta^*}{da}\frac{\partial a}{\partial \omega} < 0.$$

Hypothesis 1 also holds under the alternative increasing and convex cost function.

For Hypothesis 3, additional restrictions are required. Assume that G is three times continuously differentiable. Since

$$\frac{\partial \beta^*}{\partial \lambda} = \beta_q q^*_\lambda,$$

differentiating with respect to perceived prior variance gives

$$\frac{d}{da}\left(\frac{\partial \beta^*}{\partial \lambda}\right) = \beta_{qa} q^*_\lambda + \beta_{qq} q^*_a q^*_\lambda + \beta_q q^*_{\lambda a}. \tag{33}$$

The first two derivatives are

$$\beta_{qa} = \frac{1 + q(v_\epsilon - a)}{D(q, a)^3}, \qquad \beta_{qq} = -\frac{2a(a + v_\epsilon)}{D(q, a)^3} < 0.$$

Moreover,

$$q^*_{\lambda a} = -\frac{G''(q^*)q^*_a}{F_q} + \frac{G'(q^*)}{F_q^2}\left(F_{qa} + F_{qq}q^*_a\right), \tag{34}$$

where

$$F_{qa} = -H_{qa} > 0, \qquad F_{qq} = \lambda G'''(q^*) - H_{qq}.$$

In particular,

$$H_{qa} = -\frac{2\phi a\left[3a + 2v_\epsilon + 2qv_\epsilon(a + v_\epsilon)\right]}{D(q,a)^4} < 0,$$
$$H_{qq} = \frac{6\phi a^2(a + v_\epsilon)^2}{D(q,a)^4} > 0.$$

Equation (33) shows why convexity alone does not determine the amplification effect. The first term depends on the relative size of perceived prior variance and total signal noise, while the last term depends on higher-order properties of the cost function through $q^*_{\lambda a}$.

One transparent sufficient set of conditions is

$$a \geq v_\epsilon + (q^*)^{-1} \qquad \text{and} \qquad q^*_{\lambda a} \geq 0. \tag{35}$$

The first condition is equivalent to $\beta^* \geq 1/2$: perceived prior variance is at least as large as effective signal-noise variance, so the processed signal receives at least one-half weight. The second condition requires greater perceived prior uncertainty to make the negative response of optimal processing precision to costs less severe. This is a substantive cross-comparative-static restriction rather than a consequence of convexity alone. The first condition implies $\beta_{qa} \leq 0$. Since

$$q^*_\lambda < 0, \qquad q^*_a > 0, \qquad \beta_{qq} < 0, \qquad \beta_q > 0,$$

both of the first two terms in (33) are then non-negative, with the second strictly positive, while the second condition makes the final term non-negative. Therefore,

$$\frac{d}{da}\left(\frac{\partial\beta^*}{\partial\lambda}\right) > 0.$$

Because $\partial a/\partial\omega < 0$, this implies

$$\frac{\partial}{\partial\omega}\left(\frac{\partial\beta^*}{\partial\lambda}\right) < 0,$$

which is the amplification effect in Hypothesis 3.

Using equation (34), the second condition in (35) can equivalently be written as

$$G'(q^*)\left(F_{qa} + F_{qq}q^*_a\right) \geq G''(q^*)q^*_a F_q.$$

This makes explicit that preservation of Hypothesis 3 depends on the curvature of the processing-cost function as well as on the underlying signal environment.

A simple linear-cost example shows that amplification is not unique to the Shannon benchmark. Let $G(q) = q$, which is weakly convex. An interior solution requires $a > \sqrt{\lambda/\phi}$. Under this

condition, the first-order condition gives

$$q^* = \frac{a\sqrt{\phi/\lambda} - 1}{a + v_\epsilon}, \qquad \beta^* = \frac{a - \sqrt{\lambda/\phi}}{a + v_\epsilon}.$$

Therefore,

$$\frac{\partial \beta^*}{\partial \lambda} = -\frac{1}{2\sqrt{\phi\lambda}(a + v_\epsilon)} < 0, \qquad \frac{d}{da}\left(\frac{\partial \beta^*}{\partial \lambda}\right) = \frac{1}{2\sqrt{\phi\lambda}(a + v_\epsilon)^2} > 0.$$

Because $\partial a/\partial \omega < 0$, Hypothesis 3 holds throughout the interior in this non-Shannon example.

A complementary quadratic-cost example shows that convexity alone does not guarantee amplification. Let $G(q) = q^2/2$ and set $a = v_\epsilon = \phi = 1$ and $\lambda = 1/2$. The first-order condition in equation (32) has the unique interior solution $q^* = 1/2$, which implies $\beta^* = 1/4$. Evaluating equation (33) at these values gives

$$\frac{d}{da}\left(\frac{\partial \beta^*}{\partial \lambda}\right) = -\frac{3}{128} < 0.$$

Because $\partial a/\partial \omega < 0$, the corresponding overprecision–cost cross-partial is positive, reversing Hypothesis 3. The linear- and quadratic-cost examples therefore demonstrate that increasing and convex costs can support either sign.

The conditions in (35) are sufficient rather than necessary. The amplification effect may therefore hold outside this region, but increasing and convex processing costs alone do not guarantee its sign. Moreover, the cross-partial results in this subsection are local statements at an interior optimum. They do not by themselves sign a finite contrast between discrete Hard-Info and Easy-Info cost levels if the path between those levels crosses the zero-processing boundary or passes through regions in which the local sign condition changes. The first two predictions are consequently robust across both the benchmark and this alternative cost environment, whereas Hypothesis 3 is more demanding and depends on additional parameter and curvature restrictions. This motivates treating the Feedback-by-Hard-Info interaction as a distinctive test of the benchmark amplification channel rather than as a mechanical implication of subjectively rational inattention under every cost technology.

B Experimental material

B.1 Pictures

First block

Figure B.1: The Solvay Conference (1927)

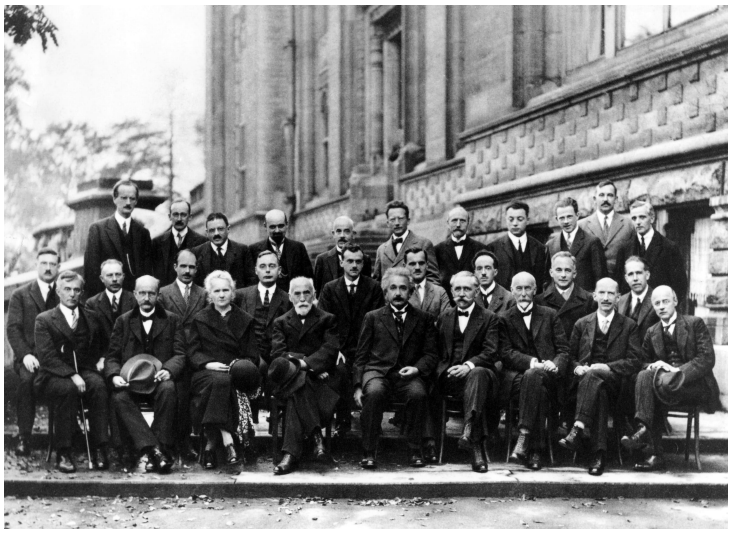


Figure B.2: Band Aid (1984)



Figure B.3: Cabinet of US President Barack Obama (2009)



Figure B.4: 75th Anniversary of Paramount (1987)



Figure B.5: Sino-German Consultations (2023)



B.1.1 Second block

Figure B.6: G20 Hamburg Summit (2017)



Figure B.7: The Irascibles (1950)



Nina Leen/Time Life Pictures/Getty Images

Life magazine's portrait of the Abstract Expressionist artists known as 'The Irascibles,' 1951. Front row: Theodore Stamos, Jimmy Ernst, Barnett Newman, James Brooks, and Mark Rothko; middle ow: Richard Pousette-Dart, William Baziotes, Jackson Pollock, Clyfford Still, Robert Motherwell, and Bradley Walker Tomlin; back row: Willem de Kooning, Adolph Gottlieb, Ad Reinhardt, and Hedda Sterne

Figure B.8: Australian Labor Party (1901)



Figure B.9: Cabinet of German Chancellor Angela Merkel (2019)



Figure B.10: British Royal Family (2007)



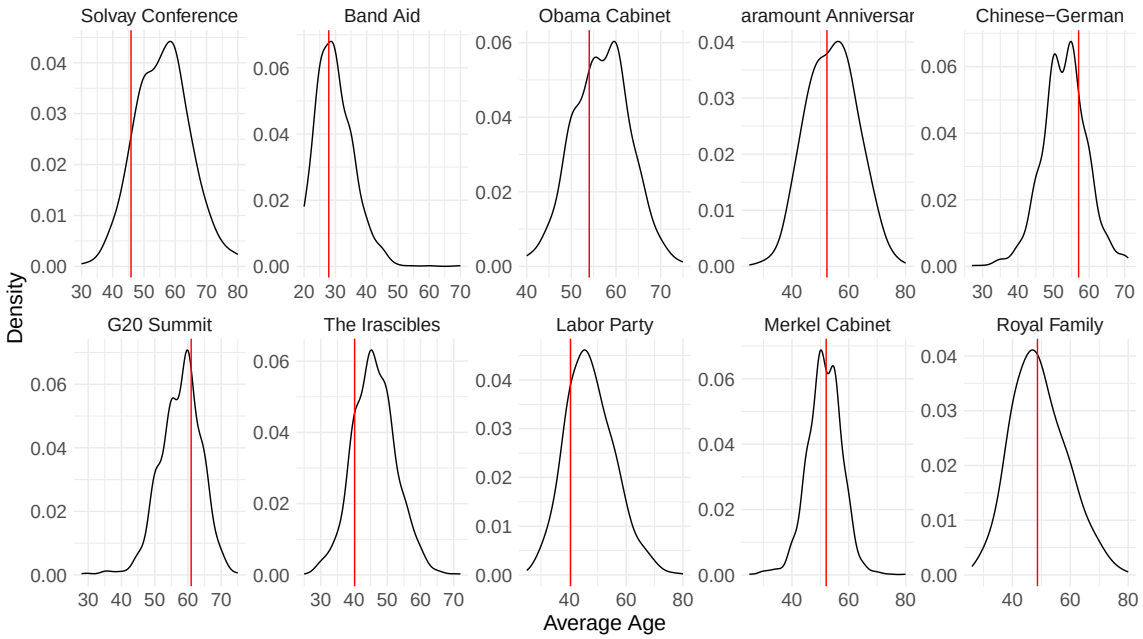
C Sample composition and descriptive figures

Table C.1: Demographic characteristics of the analysis sample

	Sample	Census[†]
Age	44.49	44.6
	(14.25)	
Female	0.504	0.507
	(0.500)	
German citizen	0.938	0.854
	(0.242)	
East Germany, including Berlin	0.196	0.194
	(0.397)	
University degree or above	0.312	0.335
	(0.463)	
N	1,200	

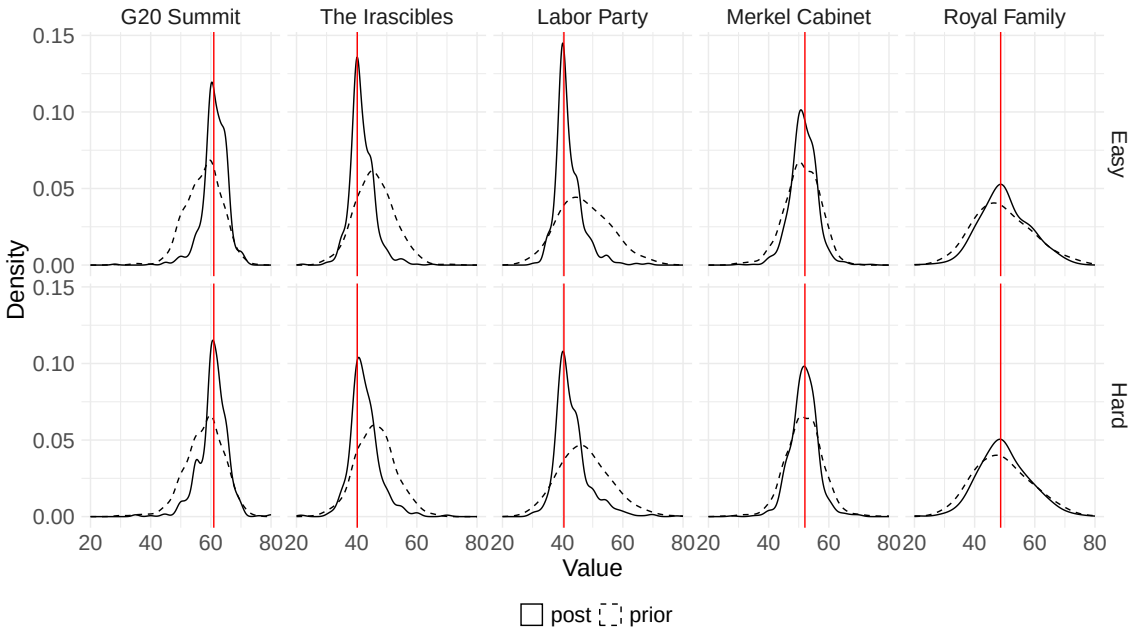
Notes: Means with standard deviations in parentheses. East Germany comprises Berlin, Brandenburg, Mecklenburg-Vorpommern, Sachsen, Sachsen-Anhalt, and Thüringen. Destatis adult-population benchmarks refer to 2019 for education and 2022 otherwise. Missing responses are excluded by variable; available observations are 1,200 for age, 1,192 for female, 1,200 for citizenship, 1,200 for East Germany, and 1,196 for education.

Figure C.1: Density of prior beliefs for each picture



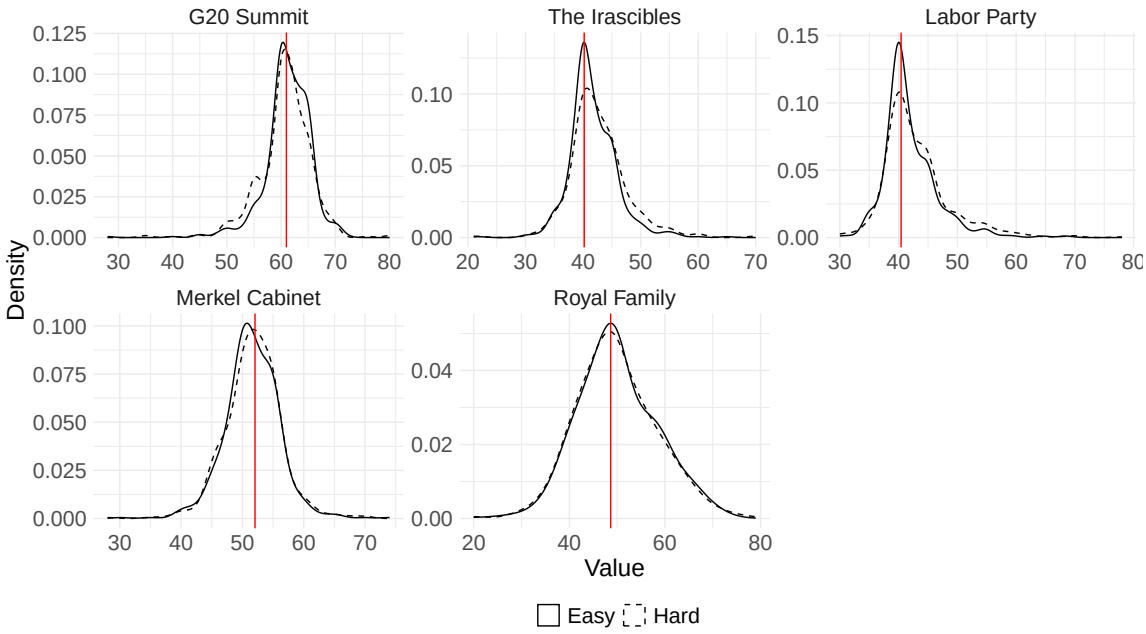
Notes: The figure plots the density of prior beliefs for each picture. The vertical red line marks the correct average age in the picture. The vertical axis is allowed to differ across pictures to make within-picture dispersion easier to see.

Figure C.2: Density of answers for each picture: common vertical scale



Notes: Each column corresponds to one Block 2 picture. The upper row shows Easy-Info and the lower row shows Hard-Info. Solid lines show posterior-belief densities; dashed lines show prior-belief densities. Vertical red lines mark the correct average ages. All panels use a common vertical scale.

Figure C.3: Density of answers for each picture: picture-specific vertical scales



Notes: Each panel shows posterior-belief densities for one Block 2 picture. Solid lines show Easy-Info and dashed lines show Hard-Info. Vertical red lines mark the correct average ages. Horizontal and vertical axis scales vary across pictures.

D Pre-registered analyses and robustness

This appendix documents the pre-registered analyses underlying the treatment results in Section 4 and reports additional robustness checks. Section D.1 reproduces the pre-registered Feedback analysis and assesses the matched comparison groups used in the main text. Section D.2 reports the pre-registered belief-updating specifications and inference. Section D.3 examines robustness to the exclusion rules and alternative data-processing choices.

D.1 Feedback manipulation checks

The pre-registered Feedback analysis compared each Feedback arm with the full No-Feedback group and used the change in average absolute overprecision from Block 1 to Block 2 as its outcome. Table D.1 reproduces this analysis. The estimates indicate movement in the intended direction in both Feedback arms relative to the pooled No-Feedback group.

Figure D.1 provides the corresponding picture-order-specific evidence. For each Feedback arm, it plots differences in picture-level absolute overprecision relative to the full No-Feedback group at each picture order, with respondent-clustered confidence intervals. Orders 1–5 capture pre-treatment differences in calibration, while orders 6–10 capture relative differences after Feedback.

Table D.1: Original full-control Feedback benchmark

	Change in absolute overprecision
Overprecise Feedback	-1.163***
	(0.259)
Underprecise Feedback	1.479***
	(0.263)
Respondents	1,200
Equal absolute magnitudes: p-value	0.490
R^2	0.077

Notes: Outcome: Block 2 minus Block 1 average absolute overprecision (actual minus subjective error). This pre-registered benchmark compares each Feedback arm with the pooled No-Feedback group, combining initial calibration strata. The revision-added symmetry test is $H_0 : \beta_{\text{Over}} + \beta_{\text{Under}} = 0$. No fixed effects; robust standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Because the pooled No-Feedback group combines participants who would have received different Feedback messages, Section 4.2 instead compares each Feedback arm with No-Feedback participants from the same Block 1 calibration category. Table D.2 shows no systematic covariate imbalance within either matched comparison.

D.2 Pre-registered belief-updating analyses and specification robustness

We assess the robustness of the main belief-updating results using the pre-registered specifications and additional checks that vary the comparison groups, controls, and regression specification.

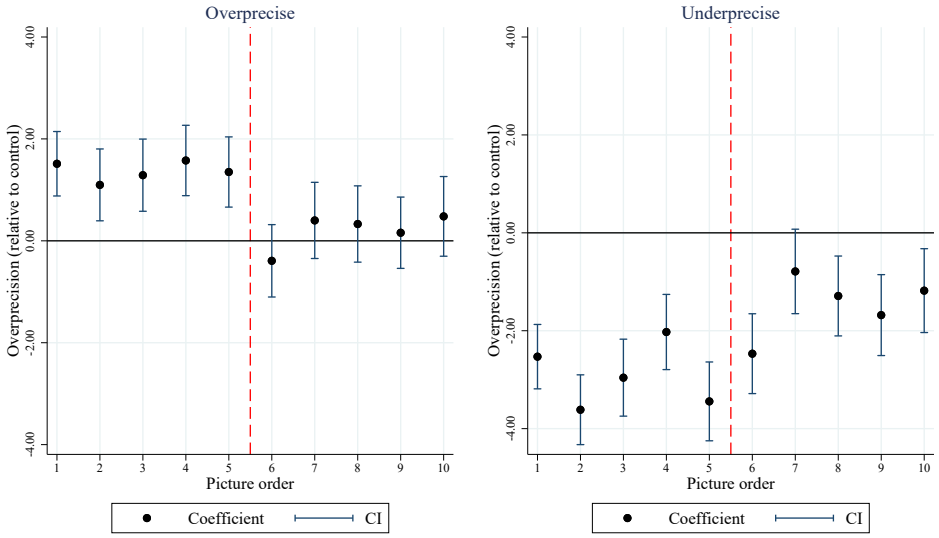
First, the pre-analysis plan specified estimating the belief-updating regression using the pooled

Table D.2: Covariate balance within matched Feedback comparison groups

	No Feedback	Feedback	Difference
Panel A: Participants classified as overprecise			
Age	44.024	44.211	0.187
			(1.110)
Female	0.522	0.475	-0.047
			(0.039)
Income	2.458	2.573	0.115
			(0.190)
University degree or above	0.292	0.319	0.027
			(0.036)
German citizen	0.956	0.931	-0.025
			(0.017)
East Germany, including Berlin	0.200	0.194	-0.006
			(0.031)
Joint balance test, p-value			0.609
Respondents	250	475	
Panel B: Participants classified as underprecise			
Age	43.377	45.819	2.443
			(1.410)
Female	0.552	0.509	-0.043
			(0.049)
Income	2.139	2.193	0.054
			(0.206)
University degree or above	0.314	0.316	0.002
			(0.046)
German citizen	0.909	0.947	0.038
			(0.026)
East Germany, including Berlin	0.201	0.193	-0.008
			(0.039)
Joint balance test, p-value			0.269
Respondents	154	321	

Notes: Panels restrict the sample according to the Block 1 classification used for the matched comparisons in the main text. The Feedback and No-Feedback columns report group means. Difference is Feedback minus No-Feedback; robust standard errors are reported in parentheses. Joint p-values come from regressions of Feedback assignment on all listed covariates. Income is represented by bracket midpoints in thousands of euros.

Figure D.1: Original full-control feedback analysis across picture order



Notes: The dependent variable is picture-level absolute overprecision, defined in equation (17). For each Feedback arm, the figure plots order-specific differences relative to the full No-Feedback group. Orders 1–5 precede Feedback and therefore represent baseline differences; orders 6–10 follow Feedback. The dashed line marks the start of Block 2. Bars are 90% confidence intervals based on standard errors clustered at the respondent level.

No-Feedback comparison group, both with and without covariates. Columns 1–2 of Table D.3 report these specifications. Columns 3–4 add the registered controls and their interactions with the signal–prior gap to the matched comparisons in Table 3. The estimates yield a similar qualitative pattern to the main results. However, the registered control set includes self-reported task difficulty, measured after treatment assignment. We therefore report the adjusted specifications in columns 2–4 for completeness and prefer the matched comparisons without these controls in the main text.

Second, the plan specified estimating the belief-updating regression separately for each Block 2 picture. Table D.4 reports these picture-specific estimates. Signal responsiveness is positive for every picture, the estimated Hard-Info effect is negative in all five pictures, and the Feedback-by-Hard-Info interactions show no systematic pattern across pictures.

Third, the plan specified an alternative formulation in which posterior beliefs are regressed on prior beliefs and their interactions with the treatments. Table D.5 reports this specification, including the corresponding treatment main effects and interactions. The results again show greater persistence of prior beliefs under Hard-Info and do not provide support for the predicted amplification effect.

Table D.6 examines two additional functional-form restrictions in the belief-updating specification. It includes the complete set of lower-order treatment intercepts associated with the slope interactions. In the pooled-control comparison, it also allows the baseline signal–prior slope and the Hard-Info slope to differ between the pre-treatment calibration strata, rather than imposing common slopes across the pooled No-Feedback group. The matched columns apply the same fully interacted specification within each calibration stratum. Hard-Info remains

Table D.3: Pre-registered and covariate-adjusted belief-updating estimates

	Full: no controls	Full: controls	Matched: overprecise	Matched: underprecise
Signal − Prior	0.595***	0.797***	0.923***	0.690***
	(0.028)	(0.201)	(0.239)	(0.150)
Overprecise Feedback × (Signal − Prior)	0.065*	0.075**	0.072*	
	(0.037)	(0.036)	(0.040)	
Underprecise Feedback × (Signal − Prior)	-0.012	-0.005		-0.029
	(0.041)	(0.041)		(0.051)
Hard-Info × (Signal − Prior)	-0.138***	-0.131***	-0.135***	-0.146**
	(0.041)	(0.039)	(0.046)	(0.063)
Overprecise Feedback × Hard-Info × (Signal − Prior)	-0.011	-0.019	-0.018	
	(0.056)	(0.053)	(0.058)	
Underprecise Feedback × Hard-Info × (Signal − Prior)	0.080	0.078		0.081
	(0.062)	(0.059)		(0.076)
Constant	1.127***	1.706	3.141	0.030
	(0.110)	(1.590)	(2.202)	(1.027)
Observations	5,946	5,862	3,531	2,331
Respondents	1,200	1,183	713	470
R^2	0.570	0.584	0.590	0.597

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. Columns 1–2 use the pooled No-Feedback comparison group; columns 3–4 match Feedback and No-Feedback respondents by Block 1 calibration. Controls in columns 2–4 are age, gender, income, education, citizenship, region, screen size, and post-treatment task difficulty, in levels and interacted with Signal − Prior. All specifications include picture fixed effects; respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table D.4: Picture-specific belief-updating estimates

	Picture 6	Picture 7	Picture 8	Picture 9	Picture 10
Signal − Prior	0.553***	0.692***	0.797***	0.648***	0.386***
	(0.060)	(0.044)	(0.048)	(0.064)	(0.047)
Overprecise Feedback × (Signal − Prior)	0.133	0.136**	0.029	-0.051	0.034
	(0.088)	(0.062)	(0.059)	(0.082)	(0.066)
Underprecise Feedback × (Signal − Prior)	0.154*	0.069	-0.095	0.019	-0.067
	(0.082)	(0.060)	(0.070)	(0.085)	(0.070)
Hard-Info × (Signal − Prior)	-0.096	-0.090	-0.226***	-0.169**	-0.076
	(0.083)	(0.059)	(0.072)	(0.080)	(0.066)
Overprecise Feedback × Hard-Info × (Signal − Prior)	0.001	-0.154*	0.014	0.133	0.008
	(0.120)	(0.086)	(0.092)	(0.118)	(0.094)
Underprecise Feedback × Hard-Info × (Signal − Prior)	0.026	-0.007	0.123	0.086	0.117
	(0.116)	(0.093)	(0.101)	(0.122)	(0.102)
Constant	0.961***	0.138	-0.045	-0.195*	0.889***
	(0.121)	(0.125)	(0.135)	(0.107)	(0.165)
Observations	1,193	1,184	1,190	1,189	1,190
Respondents	1,193	1,184	1,190	1,189	1,190
R^2	0.503	0.563	0.648	0.484	0.269

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. Each picture-specific regression estimates equation (21) using the pooled No-Feedback comparison group. No picture fixed effects; respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table D.5: Alternative belief-updating specification

	posterior
Prior	0.417***
	(0.024)
Overprecise Feedback	1.286
	(1.497)
Underprecise Feedback	-1.837
	(1.521)
Hard-Info	-3.806**
	(1.629)
Overprecise Feedback × Hard-Info	0.157
	(2.220)
Underprecise Feedback × Hard-Info	4.383*
	(2.247)
Overprecise Feedback × Prior	-0.030
	(0.030)
Underprecise Feedback × Prior	0.036
	(0.031)
Hard-Info × Prior	0.079**
	(0.033)
Overprecise Feedback × Hard-Info × Prior	-0.002
	(0.045)
Underprecise Feedback × Hard-Info × Prior	-0.090**
	(0.045)
Constant	36.672***
	(1.265)
Observations	5,946
Respondents	1,200
R^2	0.743

Notes: Outcome: posterior age estimate. The specification includes all treatment-by-prior interactions and lower-order terms, uses the pooled No-Feedback comparison group, and includes picture fixed effects. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

negatively associated with signal responsiveness, and both Feedback-by-Hard-Info interactions remain statistically insignificant and opposite to the benchmark signs. The Overprecise Feedback slope remains positive but smaller and is estimated imprecisely once the treatment intercepts are included.

Table D.6: Belief updating with saturated lower-order terms

	Full control	Matched: overprecise	Matched: underprecise
Signal − Prior	0.613***	0.595***	0.612***
	(0.037)	(0.037)	(0.038)
Initially overprecise × (Signal − Prior)	-0.019		
	(0.052)		
Overprecise Feedback × (Signal − Prior)	0.066	0.066	
	(0.045)	(0.045)	
Underprecise Feedback × (Signal − Prior)	-0.029		-0.030
	(0.048)		(0.048)
Hard-Info × (Signal − Prior)	-0.165**	-0.131**	-0.165**
	(0.065)	(0.052)	(0.065)
Initially overprecise × Hard-Info × (Signal − Prior)	0.034		
	(0.083)		
Overprecise Feedback × Hard-Info × (Signal − Prior)	-0.019	-0.019	
	(0.064)	(0.064)	
Underprecise Feedback × Hard-Info × (Signal − Prior)	0.104		0.105
	(0.080)		(0.080)
Constant	0.820***	1.332***	0.984***
	(0.270)	(0.239)	(0.278)
Observations	5,946	3,590	2,356
Respondents	1,200	725	475
R^2	0.570	0.570	0.573

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. All specifications include the lower-order treatment intercepts implied by the slope interactions and picture fixed effects. Column 1 pools No-Feedback respondents while allowing intercepts, Signal − Prior slopes, and Hard-Info effects to differ by Block 1 calibration stratum. Columns 2–3 match Feedback and No-Feedback respondents within each stratum. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The pre-analysis plan also specified directional one-sided tests, whereas the coefficient tables throughout the paper report two-sided inference. Table D.7 reports one-sided p-values for the pre-registered pooled-control specifications in columns 1–2 of Table D.3.

Table D.7: Directional inference for the pre-registered updating specifications

	Direction	Without controls	With registered controls
Overprecise Feedback × (Signal − Prior)	> 0	0.041	0.018
Underprecise Feedback × (Signal − Prior)	< 0	0.388	0.447
Hard-Info × (Signal − Prior)	< 0	< 0.001	< 0.001
Overprecise Feedback × Hard-Info × (Signal − Prior)	> 0	0.580	0.637
Underprecise Feedback × Hard-Info × (Signal − Prior)	< 0	0.902	0.907
Observations		5,946	5,862
Respondents		1,200	1,183

Notes: Entries are one-sided p-values for the pooled No-Feedback specifications in columns 1–2 of Table D.3. The pre-analysis plan specified one-sided tests but not arm-specific Feedback directions; the reported directions follow the model-to-treatment mapping in Section 4.3. Both specifications include picture fixed effects and respondent-clustered standard errors. The coefficient tables report two-sided inference.

D.3 Robustness to exclusions and data processing

We next examine whether the belief-updating results depend on the registered exclusion rules or on the processing of prior and posterior responses. Because several restrictions use responses or completion times observed after assignment, pre-registration alone does not make the restricted analysis an intention-to-treat comparison. The specification retaining all completed surveys is therefore closer to an unrestricted intention-to-treat analysis among completers.

Table D.8 re-estimates the full-control belief-updating specification under alternative sample constructions. Dropping the first-1,200 cutoff or the speeder restriction leaves the estimates broadly unchanged. Retaining all completed surveys has more noticeable effects, particularly by making the Hard-Info coefficient smaller and less precisely estimated when infeasible and extreme responses are included. The first-1,200 cutoff and speeder rule therefore do not appear to drive the main results, while the plausibility restrictions matter more for the magnitude and precision of the direct Hard-Info effect. Across these samples, the Feedback-by-Hard-Info interactions remain statistically insignificant and do not support Hypothesis 3.

Table D.9 examines the role of the pre-specified 20–80 restriction on prior and posterior responses. It keeps the retained respondent sample fixed but replaces the bounded prior and posterior variables used in the main analysis with the corresponding raw responses, thereby restoring additional picture-level observations for some retained participants. The direct Overprecise Feedback effect remains positive, while both Feedback-by-Hard-Info interactions remain statistically insignificant. The Hard-Info effect becomes smaller and less precisely estimated with the raw responses.[11]

Table D.10 implements the three additional response restrictions specified in the pre-analysis plan: restricting reports to the range of ages shown in each picture, removing reports more than three standard deviations from the within-picture median, and removing reports outside the within-picture 1st and 99th percentiles. The direct Hard-Info effect remains negative and statistically significant under each restriction. Both Feedback-by-Hard-Info interactions remain

[11] Using raw responses restores 54 picture-level observations, 51 of which contain an age report between 0 and 15 years. Exploratory influence diagnostics show that a few unusually low priors followed by large upward revisions account for much of the attenuation of the Hard-Info estimate. These diagnostics identify influential response patterns but do not establish why participants reported these values.

Table D.8: Exclusion and first-1,200 cutoff robustness

	Baseline (1,200)	No 1,200 cut	No cutoff or speeder filter	No cutoff or exclusions
Signal − Prior	0.595***	0.599***	0.597***	0.629***
	(0.028)	(0.027)	(0.027)	(0.036)
Overprecise Feedback × (Signal − Prior)	0.065*	0.070*	0.067*	0.057
	(0.037)	(0.037)	(0.037)	(0.052)
Underprecise Feedback × (Signal − Prior)	-0.012	-0.011	-0.007	0.041
	(0.041)	(0.039)	(0.039)	(0.050)
Hard-Info × (Signal − Prior)	-0.138***	-0.128***	-0.126***	-0.064
	(0.041)	(0.040)	(0.040)	(0.062)
Overprecise Feedback × Hard-Info × (Signal − Prior)	-0.011	-0.007	-0.004	0.043
	(0.056)	(0.056)	(0.056)	(0.081)
Underprecise Feedback × Hard-Info × (Signal − Prior)	0.080	0.067	0.065	0.003
	(0.062)	(0.060)	(0.060)	(0.085)
Constant	1.127***	1.142***	1.125***	-0.436
	(0.110)	(0.107)	(0.107)	(0.267)
Observations	5,946	6,285	6,311	6,605
Respondents	1,200	1,268	1,274	1,321
R^2	0.570	0.574	0.572	0.337

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. Columns 2–4 cumulatively remove the first-1,200 cutoff, speeder filter, and remaining post-completion exclusions. All estimate equation (21) with the pooled No-Feedback comparison group and picture fixed effects. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table D.9: Sensitivity of belief-updating estimates to raw prior and posterior responses

	Bounded responses	Raw responses
Signal − Prior	0.595***	0.606***
	(0.028)	(0.034)
Overprecise × (Signal − Prior)	0.065*	0.076*
	(0.037)	(0.043)
Underprecise × (Signal − Prior)	-0.012	0.008
	(0.041)	(0.046)
Hard-Info × (Signal − Prior)	-0.138***	-0.079
	(0.041)	(0.051)
Overprecise × Hard-Info × (Signal − Prior)	-0.011	-0.023
	(0.056)	(0.066)
Underprecise × Hard-Info × (Signal − Prior)	0.080	0.031
	(0.062)	(0.074)
Observations	5,946	6,000
Respondents	1,200	1,200
R^2	0.570	0.480

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. Overprecise and Underprecise indicate Feedback receipt. Bounded responses exclude reports outside 20–80; raw responses may restore picture-level observations for the same retained respondents. Both specifications use the pooled No-Feedback comparison group and picture fixed effects. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

statistically insignificant and retain signs opposite to the benchmark predictions.

Table D.10: Sensitivity to the registered response restrictions

	Main sample	Picture range	Three SDs	1st–99th pct.
Signal − Prior	0.595***	0.513***	0.622***	0.624***
	(0.028)	(0.028)	(0.027)	(0.027)
Overprecise Feedback × (Signal − Prior)	0.065*	0.088**	0.064*	0.062*
	(0.037)	(0.037)	(0.033)	(0.034)
Underprecise Feedback × (Signal − Prior)	-0.012	0.020	-0.032	-0.028
	(0.041)	(0.040)	(0.036)	(0.037)
Hard-Info × (Signal − Prior)	-0.138***	-0.090**	-0.112***	-0.118***
	(0.041)	(0.039)	(0.037)	(0.038)
Overprecise Feedback × Hard-Info × (Signal − Prior)	-0.011	-0.019	-0.022	-0.025
	(0.056)	(0.054)	(0.048)	(0.049)
Underprecise Feedback × Hard-Info × (Signal − Prior)	0.080	0.041	0.056	0.049
	(0.062)	(0.059)	(0.053)	(0.054)
Observations	5,946	5,389	5,839	5,809
Respondents	1,200	1,200	1,200	1,200
R^2	0.570	0.490	0.609	0.604

Notes: Outcome: posterior minus prior; Signal is the displayed mean age. Column 1 omits reports outside 20–80. Columns 2–4 separately add exclusions for reports outside the picture's age range, more than three standard deviations from the within-picture median, or outside the within-picture 1st–99th percentiles. Restrictions apply separately to prior and posterior reports. All use the pooled No-Feedback comparison group and picture fixed effects; respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table D.11 directly examines whether the post-completion exclusions differ by randomized assignment. Within both pre-treatment calibration groups, exclusion rates are virtually identical between Feedback and No-Feedback participants. The rate is somewhat higher under Hard-Info than under Easy-Info, but the difference is statistically insignificant. A joint test across the treatment assignments, calibration stratum, and their interactions also fails to reject equal exclusion rates ($p = 0.627$). The all-completers estimates in Table D.8 provide a direct check of sensitivity to these exclusions.

Table D.11: Post-completion exclusion rates by randomized assignment

Comparison	Control	Treatment	Difference	p-value
Feedback, initially overprecise	4.4	4.2	-0.2	0.901
Feedback, initially underprecise	3.6	3.6	0.1	0.968
Information complexity	3.3	4.7	1.4	0.201

Notes: Exclusion rates (percent) among all 1,321 completers before the first-1,200 cutoff. Rows 1–2 compare Feedback with No-Feedback within initial calibration strata; row 3 compares Hard-Info with Easy-Info. Differences are Treatment minus Control, in percentage points. Exclusions combine registered subjective-error feasibility, response-range, and speeder rules. p-values use heteroskedasticity-robust linear probability models; the joint test of treatment assignments, calibration stratum, and all interactions gives $p = 0.627$.

E Additional evidence on the information-complexity manipulation

Table E.1: Information treatment and time spent processing the information

	OLS	Median	Robust
Hard-Info	0.716	1.426***	1.286***
	(1.144)	(0.439)	(0.333)
Constant	18.078***	10.151***	10.451***
	(1.116)	(0.377)	(0.306)
Observations	6,000	6,000	6,000
Respondents	1,200	1,200	1,200
R^2	0.002		0.010
Pseudo-R^2		0.003	

Notes: The outcome is seconds spent on the information page. All models include picture fixed effects. Standard errors in parentheses are respondent-clustered for OLS and use a respondent-clustered bootstrap with 999 replications for median and robust regression. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table E.2: Feedback effects on processing time by information condition

	Initially overprecise	Initially underprecise
Feedback effect, Easy-Info	0.053	0.117
	(0.079)	(0.097)
Feedback effect, Hard-Info	0.153*	0.126
	(0.081)	(0.092)
Difference: Hard-Info − Easy-Info	0.099	0.009
	(0.113)	(0.134)
Observations	3,625	2,375
Respondents	725	475
R^2	0.009	0.011

Notes: The outcome is log time spent on the signal page. Feedback effects compare Feedback with No-Feedback within the indicated Block 1 calibration group. The difference is the Feedback × Hard-Info interaction. Both models include picture fixed effects. Respondent-clustered standard errors are in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table E.3: How Hard-Info changes belief-revision behavior

Outcome	Unit	Hard-Info effect (SE)
Exact zero revision	Percentage points	3.673***(1.274)
Wrong-way revision relative to displayed signal	Percentage points	1.163(0.944)
Absolute belief revision	Years	−0.735***(0.153)
Posterior more accurate than prior	Percentage points	−2.874**(1.385)
Posterior absolute error	Years	0.491***(0.143)
Held-out absolute residual	Years	−0.130(0.102)
Held-out squared residual	Years squared	−1.849(1.510)
Observations		5,946
Respondents		1,200

Notes: Each row regresses the listed outcome on Hard-Info with picture fixed effects and respondent-clustered standard errors. Wrong-way revisions move away from the displayed signal average. Held-out residuals are prediction errors on a fifth picture after estimating treatment-specific intercepts and signal-response slopes on the other four; they measure dispersion after removing treatment-specific mean updating. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

F Diagnostics and robustness

This appendix reports the additional analyses discussed in Section 5.

F.1 Statistical explanations

A first possibility is that the pre-registered interaction is statistically difficult to detect. We therefore simulate the sensitivity of the full-control-group, no-covariate specification in column (1) of Appendix Table D.3. The simulation preserves the five-picture panel and respondent-level dependence, using 2,000 respondent-level wild-bootstrap replications and two-sided 5 percent tests. Effects are injected in the directions implied by the model: positive for Overprecise Feedback × Hard-Info and negative for Underprecise Feedback × Hard-Info.

Table F.1: Power of the pre-registered interaction test

Interaction magnitude	Overprecise Feedback	Underprecise Feedback
0.05	0.148	0.130
0.10	0.420	0.379
0.15	0.746	0.680
0.20	0.938	0.898

Notes: Entries are simulated rejection rates for two-sided 5 percent tests. Interaction magnitude is shown in absolute value. Null rejection rates are 0.059 and 0.057.

Power is below one-half for interaction magnitudes of 0.10 and reaches about 80 percent only at magnitudes of roughly 0.16 and 0.18. Thus, conditional on the registered interaction being the correct test of the mechanism, the design has limited power against small-to-moderate effects. Low power also makes sign reversals due to sampling variation more likely. The opposite-signed point estimates consequently provide no affirmative evidence for the benchmark prediction, but neither their signs nor their statistical insignificance sharply distinguish a small interaction from zero.

A second related statistical concern is that the pre-registered amplification test places substantial demands on the precision of reported beliefs. This test relies on three-way interactions between the signal–prior gap and the two treatments. Because the reported prior enters both belief revision and the signal–prior gap, rounding or heaping can introduce measurement noise directly into the terms identifying these effects. Table F.2 therefore considers three progressively simpler reduced-form specifications that retain the randomized treatment indicators while removing the signal–prior interaction structure. Column 1 retains belief revision as the dependent variable but removes the signal–prior gap and its interactions. Column 2 instead uses posterior absolute error, which depends on the reported posterior and the fixed true age but not on the reported prior. Column 3 augments this specification with respondents' average actual error in Block 1, measured before Feedback is provided, to account for persistent differences in baseline forecasting accuracy without conditioning on a post-treatment variable. We interpret these specifications as robustness checks.

Table F.2: Reduced-form treatment effects under alternative outcome specifications

	Revision	Posterior error	Posterior error
Hard-Info	0.042	0.578**	0.561**
	(0.288)	(0.246)	(0.232)
Overprecise Feedback	-0.239	-0.104	-0.412*
	(0.319)	(0.221)	(0.218)
Underprecise Feedback	0.242	-0.411*	0.005
	(0.282)	(0.225)	(0.216)
Overprecise Feedback × Hard-Info	0.348	-0.028	0.083
	(0.421)	(0.347)	(0.335)
Underprecise Feedback × Hard-Info	-0.461	-0.319	-0.249
	(0.394)	(0.338)	(0.321)
Average actual error (Block 1)			0.268***
			(0.033)
Constant	3.201***	3.101***	1.220***
	(0.248)	(0.175)	(0.263)
Observations	5,946	5,946	5,946
Respondents	1,200	1,200	1,200
R^2	0.194	0.090	0.115

Notes: The outcomes are belief revision (posterior minus prior) in column 1 and posterior absolute error (absolute deviation from the rounded task benchmark) in columns 2–3. Column 3 adds average actual error across the five pre-Feedback Block 1 pictures. All columns use the same Block 2 sample and pooled No-Feedback comparisons, with picture fixed effects and respondent-clustered standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

F.2 Theoretical explanations

The benchmark model allows two forms of behavior that require care when mapping observed revisions to the updating rule. First, exact zero revisions may reflect the model's zero-attention corner, although coarse reporting or rounding can generate the same observed behavior. Second, participants may sometimes revise their beliefs away from the displayed signal. Such wrong-way revisions can arise within the model because participants respond to the signal they perceive after processing noise rather than directly to the displayed signal.

To see this, let

$$g_{i,k} = signal_k - prior_{i,k}$$

denote the objective signal–prior gap. Conditional on an interior update rate, a wrong-way revision occurs when processing noise is large enough to push the perceived signal to the opposite side of the prior from the displayed signal. Under the Gaussian processing-noise assumption,

$$\Pr(\text{wrong-way} \mid g_{i,k}) = \Phi\left(-\frac{|g_{i,k}|}{\sigma_{\psi,i}}\right).$$

Wrong-way revisions are therefore compatible with the benchmark model and become more likely as processing noise increases relative to the signal–prior gap. They are not, however, unique to this mechanism, since biased signal interpretation or other non-Bayesian updating rules can generate the same behavior.

Table F.3 therefore re-estimates the treatment specification after first excluding exact zero

revisions and then, more restrictively, excluding revisions that move away from the displayed signal. Neither restriction recovers the predicted Feedback-by-Hard-Info interaction.

Table F.3: Treatment interactions under response-pattern restrictions

	Nonzero revision	Toward signal
Signal − Prior	0.660***	0.731***
	(0.029)	(0.028)
Overprecise × (Signal − Prior)	0.043	0.065*
	(0.038)	(0.035)
Underprecise × (Signal − Prior)	-0.012	-0.004
	(0.038)	(0.036)
Hard-Info × (Signal − Prior)	-0.120***	-0.112***
	(0.041)	(0.041)
Overprecise × Hard-Info × (Signal − Prior)	0.004	-0.001
	(0.056)	(0.052)
Underprecise × Hard-Info × (Signal − Prior)	0.079	0.084
	(0.060)	(0.054)
Constant	1.241***	0.992***
	(0.124)	(0.116)
Observations	4,823	4,070
Respondents	1,191	1,178
R^2	0.631	0.763

Notes: The outcome is Posterior − Prior; Signal is the displayed average age. Column 1 excludes zero revisions. Column 2 additionally requires $(Posterior - Prior)(Signal - Prior) > 0$, excluding wrong-way revisions but allowing overshooting. Both columns use pooled No-Feedback comparisons, picture fixed effects, and respondent-clustered standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

F.3 Design-based explanations

A design-based explanation for the unsupported treatment interaction is that the Feedback manipulation may affect perceived processing costs in addition to perceived prior precision. To see how such a spillover affects the interpretation of the crossed treatment coefficient, consider a local approximation to the latent update rate,

$$\beta_{i,k} \approx a + b\omega_i + c\lambda_k + d\omega_i\lambda_k,$$

where $b < 0$, $c < 0$, and $d < 0$. Suppose first that Feedback changes only perceived prior precision by $\Delta\omega_F$. The difference between the Feedback effect under Hard-Info and its effect under Easy-Info is then proportional to

$$d\Delta\omega_F(\lambda_H - \lambda_E),$$

which corresponds to the cross-partial underlying Hypothesis 3.

Now suppose that Feedback also changes perceived processing costs, and allow this spillover to differ between Hard-Info and Easy-Info. Denote these changes by $\Delta\lambda_{F,H}$ and $\Delta\lambda_{F,E}$. The Hard-minus-Easy difference in the Feedback effect then additionally contains

$$(c + d\omega_i + d\Delta\omega_F)(\Delta\lambda_{F,H} - \Delta\lambda_{F,E}).$$

Thus, the crossed treatment coefficient isolates the theoretical amplification effect only if Feedback does not differentially change perceived processing costs across the two information treatments. More generally, interpreting the experimental interaction as the model's ceteris-paribus cross-partial requires Feedback to shift perceived prior precision without also changing other determinants of information processing, and Hard-Info to shift processing costs without changing perceived prior precision or signal content.

Table F.4 provides a diagnostic for this mapping condition using self-reported difficulty, measured in the post-experimental questionnaire. Respondents rated how difficult it was to view the pictures and information clouds on a scale from 1 (not at all difficult) to 5 (very difficult). We regress reported difficulty on the Feedback and information-treatment assignments and their interactions. Overprecise Feedback is associated with greater reported difficulty, while the Feedback-by-Hard-Info interactions are negative but imprecisely estimated. Because the difficulty item is broad and measured after treatment, it does not directly measure the model's processing-cost parameter λ. We therefore do not interpret perceived difficulty as an explanation for the null interaction. Instead, the exercise illustrates that Feedback may shift beliefs about the task beyond perceived prior precision, so the crossed treatment coefficient should be interpreted as an interaction between two randomized interventions rather than as a structural estimate of the theoretical cross-partial.

Table F.4: Self-reported task difficulty by treatment assignment

	Baseline	With controls
Hard-Info	0.189*	0.182
	(0.114)	(0.116)
Overprecise Feedback	0.344***	0.326***
	(0.113)	(0.114)
Underprecise Feedback	-0.023	0.001
	(0.106)	(0.108)
Overprecise Feedback × Hard-Info	-0.201	-0.176
	(0.160)	(0.162)
Underprecise Feedback × Hard-Info	-0.180	-0.178
	(0.159)	(0.162)
Constant	2.059***	2.510***
	(0.077)	(0.596)
Respondents	1,195	1,183
R^2	0.020	0.051

Notes: The outcome is post-experimental difficulty of viewing pictures and signal displays (1–5). Column 2 controls for age, gender, income, education, German citizenship, region, and logged screen resolution (megapixels). These are respondent-level regressions with heteroskedasticity-robust standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

F.4 Exploratory analysis using measured overprecision

We examine whether responsiveness to Hard-Info varies with measured overprecision. Instead of using Feedback assignment to shift perceived prior precision, this specification uses respondents'

measured overprecision directly and interacts it with the randomized Hard-Info treatment. This avoids relying on the Feedback first stage, but measured overprecision is not randomly assigned. The resulting interaction is therefore correlational rather than a causal effect of overprecision.

We measure effective signal responsiveness using the update ratio,

$$U_{i,k} = \frac{posterior_{i,k} - prior_{i,k}}{signal_k - prior_{i,k}}, \tag{36}$$

which measures the fraction of the signal–prior gap closed by the posterior. Using equation (22),

$$U_{i,k} = \beta_{i,k} + \frac{\beta_{i,k}\psi_{i,k}}{signal_k - prior_{i,k}}.$$

Under the same mean-zero independence assumption used in Section 4.3, the update ratio therefore measures the underlying update rate in expectation. A limitation is that the ratio can become unstable when the displayed signal lies close to the prior, because the signal–prior gap appears in the denominator. We address this concern in the robustness analyses below. We estimate

$$U_{i,k} = \alpha + \gamma_0 Hard_i + \gamma_1 \overline{op}_{i,b} + \gamma_2 \overline{op}_{i,b} \times Hard_i + \eta_k + u_{i,k}, \tag{37}$$

where $\overline{op}_{i,b}$ denotes measured absolute overprecision in block b. A negative γ_2 is directionally consistent with the amplification prediction. Absolute overprecision is not, however, structurally equivalent to ω_i across respondents with heterogeneous calibrated error variances: equations (2)–(3) show that its level depends on both calibration and the scale of forecasting error. The theoretically direct index of sensitivity to processing costs is instead inverse perceived variance, which is proportional to SE_i^{-2} in equation (2). We therefore treat γ_2 as an exploratory moderation coefficient rather than a structural estimate of the overprecision–cost cross-partial.

Table F.5 reports estimates using both Block 1 and Block 2 overprecision. Block 1 is measured before Feedback, while Block 2 is closer to the updating decision but may itself be affected by Feedback. The estimated interactions are negative in both cases, although only the Block 2 estimate is statistically significant at the 10% level.

A key limitation of this specification is that overprecision itself is measured rather than randomly assigned, so variation across measured overprecision does not identify a causal effect of overprecision. This concern is particularly relevant for the Block 2 measure, which is constructed after treatment assignment. Appendix Table F.6 reports robustness checks. Column 1 reproduces the Block 2 full-sample specification from Table F.5. Column 2 adds pre-treatment demographic characteristics. Column 3 instead restricts the analysis to the No-Feedback group, in which measured Block 2 overprecision cannot itself have been shifted by the calibration-feedback treatment. Column 4 combines both restrictions. Column 5 repeats the original specification in the No-Feedback sample, including reported difficulty and its interaction with Hard-Info. Adding the pre-treatment controls leaves the interaction essentially unchanged. Restricting the sample to No-Feedback participants preserves its negative sign but produces a less precise estimate, as does combining the two checks. Including the original difficulty controls also leaves the interaction negative and statistically insignificant. These exercises suggest that observable demographic

Table F.5: Update ratio, measured overprecision, and information complexity

	Block 1	Block 2
Absolute overprecision	0.045	0.018
	(0.031)	(0.021)
Hard-Info	-0.163	-0.161
	(0.153)	(0.152)
Absolute overprecision × Hard-Info	-0.045	-0.060*
	(0.036)	(0.035)
Constant	0.710***	0.782***
	(0.100)	(0.088)
Observations	5,946	5,946
Respondents	1,200	1,200
R^2	0.003	0.003

Notes: The outcome is the update ratio (equation (36)). Columns 1 and 2 estimate equation (37) using Block 1 and Block 2 absolute overprecision, respectively. Both use the full sample, picture fixed effects, and respondent-clustered standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

differences do not drive the full-sample estimate and show that the negative sign persists without Feedback-induced variation in calibration. They do not make measured overprecision exogenous, however, and the smaller No-Feedback sample limits the precision of the comparison.

Because the update ratio divides belief revision by the signal–prior gap, its distribution can be sensitive to both the underlying response measures and observations with small denominators. Table F.7 reconstructs the ratio using raw rather than bounded prior and posterior responses while holding the measured-overprecision moderators fixed; the negative interaction is preserved and becomes somewhat larger in magnitude. Table F.8 examines alternative constructions, including denominator restrictions, winsorization, participant-level measures of responsiveness, leave-one-picture-out overprecision, and a direct revision-on-gap specification that avoids division by the signal–prior gap. The interaction generally retains its negative sign across these alternatives, although its statistical significance is not robust.

We also examine the stability of measured overprecision. Table F.9 assesses its consistency across pictures and blocks. Within-block consistency and cross-block correlation of absolute overprecision are substantially lower than those of subjective error, and the majority overprecision classification also shows limited stability across blocks. These statistics do not constitute a classical measurement-error correction, particularly because the pictures differ in difficulty, but they indicate that the composite overprecision measure contains substantial picture-specific variation.

Absolute overprecision combines actual error and subjective error. We next examine these components separately and consider relative overprecision as an alternative measure. Table F.10 reproduces the Block 2 specification in Table F.5 and then replaces average absolute overprecision with average subjective error, average actual error, and relative overprecision, one at a time. Higher subjective error indicates greater perceived prior uncertainty, so the model-consistent interaction with Hard-Info is positive in this specification. The estimated Subjective Error × Hard-Info interaction has this sign and is marginally significant, whereas the Actual Error ×

Table F.6: Robustness of the Block-2 measured-overprecision interaction

	Full sample		No-Feedback		
	Baseline	+ Demographics	Baseline	+ Demographics	Original specification
Absolute overprecision	0.018	0.023	0.025	0.026	0.022
	(0.021)	(0.021)	(0.064)	(0.061)	(0.062)
Hard-Info	-0.161	-0.162	-0.097	-0.161	-0.478
	(0.152)	(0.153)	(0.336)	(0.339)	(0.443)
Absolute overprecision × Hard-Info	-0.060*	-0.060*	-0.121	-0.110	-0.121
	(0.035)	(0.035)	(0.097)	(0.095)	(0.097)
Reported difficulty					-0.109
					(0.109)
Difficulty × Hard-Info					0.180
					(0.170)
Constant	0.782***	-0.576	0.775***	0.019	1.003***
	(0.088)	(1.060)	(0.185)	(0.847)	(0.301)
Observations	5,946	5,886	2,003	1,993	1,998
Respondents	1,200	1,188	404	402	403
R^2	0.003	0.007	0.008	0.018	0.008

Notes: The outcome is the update ratio (equation (36)), using Block 2 absolute overprecision. Columns 1–2 use the full sample; columns 3–5 use No-Feedback respondents. Columns 2 and 4 add pre-treatment controls for age, gender, income, education, German citizenship, and region. Column 5 reproduces the original specification, adding reported difficulty and its Hard-Info interaction; these post-treatment controls are diagnostic. All include picture fixed effects and respondent-clustered standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table F.7: Sensitivity of update-ratio estimates to raw prior and posterior responses

	Block 2, bounded	Block 2, raw	Block 1, bounded	Block 1, raw
Overprecision (Block 2) × Hard-Info	-0.060*	-0.089**		
	(0.035)	(0.042)		
Overprecision (Block 1) × Hard-Info			-0.045	-0.097**
			(0.036)	(0.049)
Observations	5,946	6,000	5,946	6,000
Respondents	1,200	1,200	1,200	1,200
R^2	0.003	0.002	0.003	0.002

Notes: The outcome is the picture-level update ratio. Bounded columns apply the pre-specified 20–80 prior/posterior restriction; raw columns use unrestricted responses from the same retained respondents. Overprecision moderators are unchanged. All include picture fixed effects and omit reported difficulty, with respondent-clustered standard errors in parentheses. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table F.8: Sensitivity of the measured-overprecision × Hard-Info interaction to measurement and outcome construction

Specification	Interaction estimate (SE)	p-value	Observations	Respondents
Baseline: Block-1 overprecision	$-0.045(0.036)$	0.212	5,946	1,200
Baseline: Block-2 overprecision	$-0.060^{*}(0.035)$	0.086	5,946	1,200
Exclude $\lvert\text{Signal} - \text{Prior}\rvert < 0.5$	$-0.017(0.015)$	0.246	5,542	1,200
Exclude $\lvert\text{Signal} - \text{Prior}\rvert < 1$	$-0.011(0.009)$	0.240	5,344	1,200
Winsorized at 1st/99th percentiles	$-0.028(0.017)$	0.102	5,946	1,200
Leave-one-picture-out Block-2 overprecision	$-0.058^{**}(0.028)$	0.040	5,946	1,200
Participant median update ratio	$-0.019^{*}(0.010)$	0.053	1,200	1,200
Participant through-origin signal-response slope	$-0.012^{*}(0.007)$	0.091	1,200	1,200
Direct revision-on-gap specification	$-0.006(0.009)$	0.493	5,946	1,200

Notes: Picture-level specifications use Block 2 absolute overprecision unless otherwise indicated. Leave-one-picture-out overprecision uses the other four Block 2 pictures. The participant-level slope comes from a through-origin regression of belief revision on the signal–prior gap. The final specification regresses belief revision directly on the gap and reports the Gap × Overprecision × Hard-Info coefficient; other rows report the Overprecision × Hard-Info coefficient. All specifications include the relevant lower-order terms. Picture-level regressions include picture fixed effects and respondent-clustered standard errors; participant-level regressions use heteroskedasticity-robust standard errors. Coefficient magnitudes are not directly comparable across outcomes and aggregation levels. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table F.9: Measurement stability across pictures and blocks

Measure	No-Feedback Block-1 α	No-Feedback Block-2 α	No-Feedback cross-block r
Composite overprecision	0.411	0.395	0.380
Subjective error	0.730	0.739	0.574
Majority-classification agreement		57.6 percent	
Cohen's κ		0.130	

Notes: Cronbach's α, cross-block correlations, classification agreement, and Cohen's κ use No-Feedback respondents. Complete-case counts for Block-2 alpha are 398 for composite overprecision and 403 for subjective error; classification agreement uses 382 respondents observed on all five pictures in both blocks. The majority classification labels a respondent overprecise when at least three of five picture-level measures are positive. Because pictures differ in difficulty, the statistics describe measurement stability rather than imposing a classical measurement-error model.

Hard-Info interaction is small and statistically insignificant. These estimates suggest a role for perceived uncertainty, but do not establish which component drives the interaction with absolute overprecision. Using relative overprecision preserves the negative sign of the interaction, but the estimate is imprecise.

The last two columns use the theory-aligned inverse subjective-variance index, measured as $100/\overline{SE}_{i,b}^2$ for readability. Its interaction with Hard-Info is negative using both the pre-Feedback Block 1 measure and the Block 2 measure. Both estimates have the sign implied by the benchmark model but are statistically insignificant. Overall, these exercises provide suggestive evidence of a relationship with perceived uncertainty, but limited evidence that the interaction is robust to alternative measures of precision.

Table F.10: Alternative precision moderators in the update-ratio regression

	Absolute overprecision	Subjective error	Actual error	Relative overprecision	Inverse subjective variance: Block 1	Inverse subjective variance: Block 2
Moderator	0.018	-0.058*	-0.041	0.157	0.012	0.033
	(0.021)	(0.035)	(0.030)	(0.343)	(0.008)	(0.023)
Hard-Info	-0.161	-0.787***	-0.351	-0.183	-0.175	-0.056
	(0.152)	(0.277)	(0.325)	(0.160)	(0.138)	(0.187)
Moderator × Hard-Info	-0.060*	0.104*	0.014	-0.547	-0.011	-0.035
	(0.035)	(0.054)	(0.039)	(0.460)	(0.008)	(0.023)
Observations	5,946	5,946	5,946	5,946	5,852	5,847
Respondents	1,200	1,200	1,200	1,200	1,181	1,180
R^2	0.003	0.003	0.002	0.002	0.003	0.003

Notes: Each column is a separate picture-level update-ratio regression. Column 1 reproduces the Block 2 specification in Table F.5; columns 2–4 use the labelled Block 2 moderators. Absolute and relative overprecision follow equations (18) and (19). Columns 5–6 use $100/\overline{SE}_{i,b}^2$ from Blocks 1 and 2, excluding 19 and 20 respondents with zero average subjective error, respectively. All include picture fixed effects and respondent-clustered standard errors. Moderator scales differ, so coefficient magnitudes are not directly comparable. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Finally, the measured-overprecision interaction could reflect other observable features of the information-processing environment. Table F.11 therefore re-estimates the Block 2 update-ratio specification while accounting for processing time, reported task difficulty, and signal informativeness, first separately and then jointly. For signal informativeness, we use the ratio of 15, the number of ages displayed in the information signal, to the total number of people shown in the corresponding picture; higher values therefore indicate that the signal represents a larger share of the underlying group. Because informativeness varies only across pictures, we include picture fixed effects, allowing its association with updating to differ under Hard-Info. Processing time provides an imperfect proxy for engagement with the information, while reported difficulty captures respondents' perceived difficulty in processing the visual information. The estimated Overprecision × Hard-Info interaction remains largely unchanged when these measures are included. Because processing time and reported difficulty are measured after treatment and may themselves respond to treatment assignment, these specifications are diagnostic rather than causal adjustments.

Table F.11: Observable alternative explanations for the update-ratio interaction

	Baseline	+ Time	+ Informativeness	+ Difficulty	+ All
Average overprecision (Block 2)	0.018	0.021	0.018	0.018	0.021
	(0.021)	(0.022)	(0.021)	(0.021)	(0.021)
Hard-Info	-0.164	0.114	0.042	-0.484**	-0.036
	(0.153)	(0.403)	(0.317)	(0.226)	(0.471)
Overprecision × Hard-Info	-0.060*	-0.062*	-0.060*	-0.062*	-0.064*
	(0.035)	(0.036)	(0.035)	(0.036)	(0.036)
Log processing time		0.190*			0.186*
		(0.099)			(0.101)
Log processing time × Hard-Info		-0.116			-0.099
		(0.137)			(0.136)
Signal informativeness × Hard-Info			-0.313		-0.316
			(0.546)		(0.542)
Reported task difficulty				-0.073	-0.063
				(0.070)	(0.071)
Task difficulty × Hard-Info				0.146	0.144
				(0.099)	(0.099)
Observations	5,922	5,922	5,922	5,922	5,922
Respondents	1,195	1,195	1,195	1,195	1,195
R^2	0.003	0.003	0.003	0.003	0.004

Notes: The outcome is the picture-level update ratio; all columns use Block 2 absolute overprecision and the same complete-case sample. Columns 2–5 add the labelled diagnostics and their Hard-Info interactions. Informativeness is 15 divided by the number of people pictured; its main effect is absorbed by picture fixed effects, included throughout. Respondent-clustered standard errors are in parentheses. Processing time and difficulty are post-treatment measures, so these are diagnostic rather than causal adjustments. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.